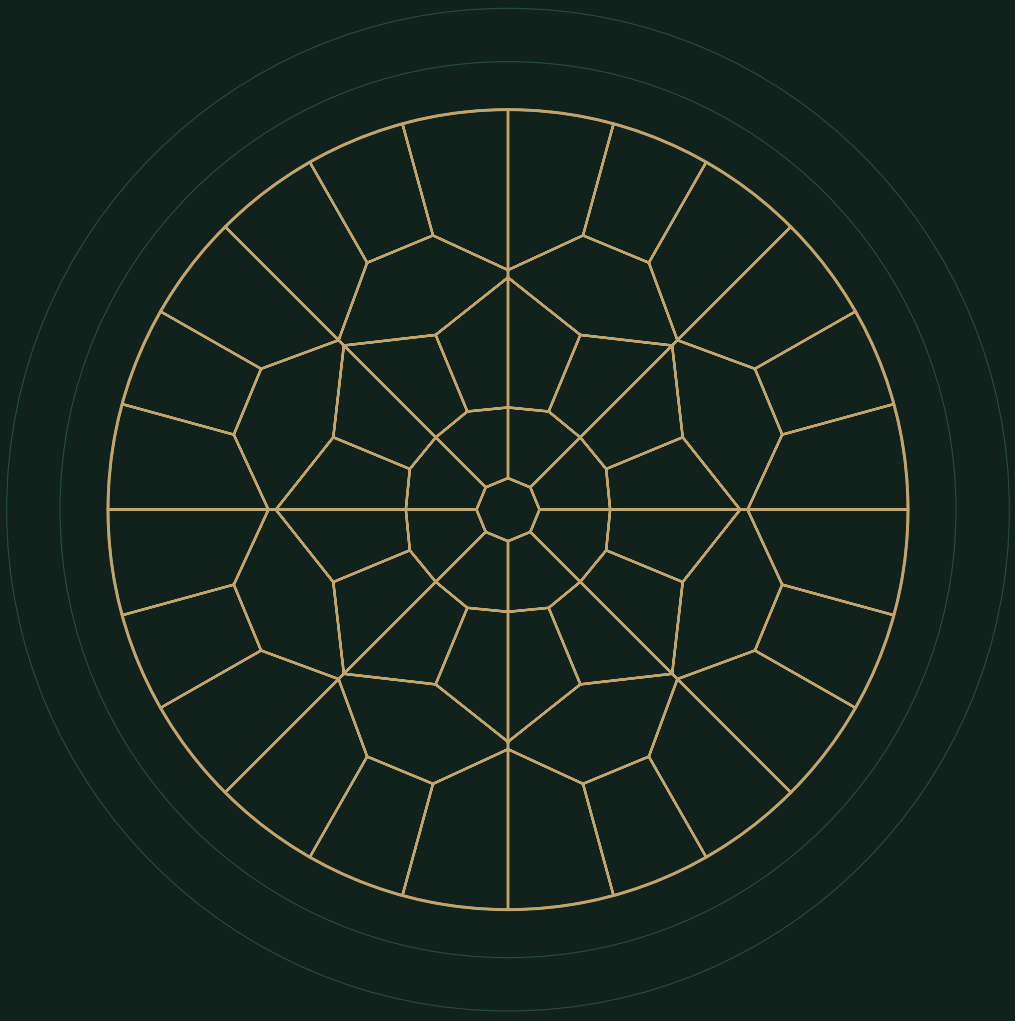
VORONOI GARDENS



A Coloring Book of Cellular Geometry

50 Voronoi Mandalas & Organic Tessellations

MathRelaxing
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VORONOI GARDENS

A Coloring Book of Cellular Geometry

50 Voronoi Mandalas & Organic Tessellations · MathRelaxing

About this book

Every page in this book begins with nothing but a scatter of points. From each point grows a cell · the region of space closer to that point than to any other. Together these cells lock into a Voronoi diagram: the same hidden geometry that shapes honeycombs, dragonfly wings, cracked riverbeds and the seeds of a sunflower.

Fifty unique designs await · radial mandalas, golden-angle spirals, cathedral rose windows and softly relaxed organic cells. Each is drawn as clean line art, every cell a closed region ready for your colour.

How to colour:

- Any pens, pencils or markers work beautifully.
- Colour each cell on its own, or blend across neighbours.
- Work from the centre outward to watch the pattern bloom.
- On an iPad, drop the artwork into GoodNotes or Procreate.

Enjoy the quiet geometry of growth.

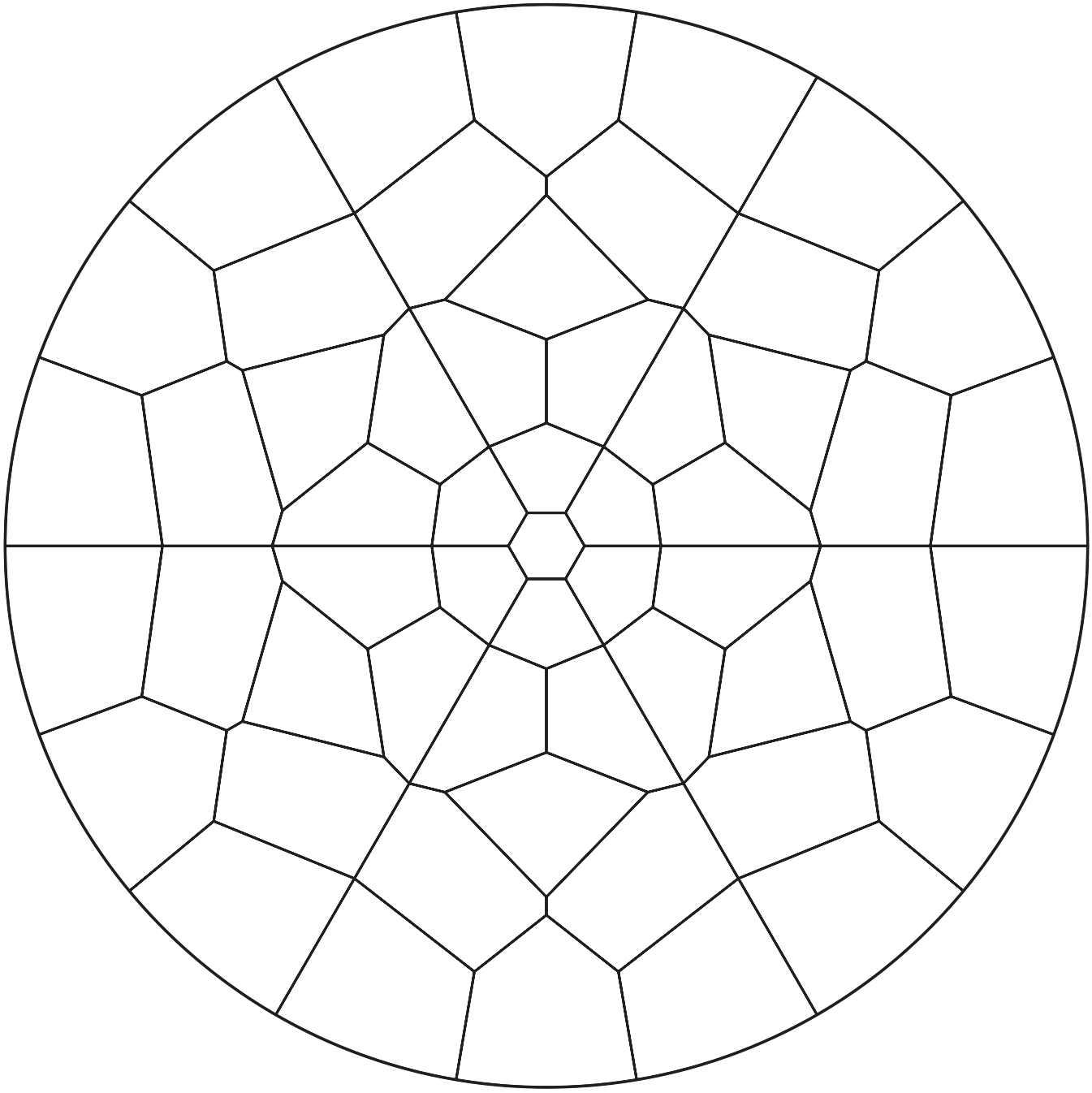
The mathematics

A Voronoi diagram is built from a set of seed points. For every location on the page we ask a single question: which seed is nearest? All the points that answer with the same seed form that seed's cell. The cell walls fall exactly halfway between neighbouring seeds.

Where the seeds sit decides everything. Place them with k-fold rotational symmetry and a mandala appears. Place them along the golden angle of 137.5° and you recreate the spiral packing of a sunflower head. Scatter them at random, then nudge each seed toward the centre of its own cell · a step called Lloyd's relaxation · and the mesh softens into organic, bubble-like tissue.

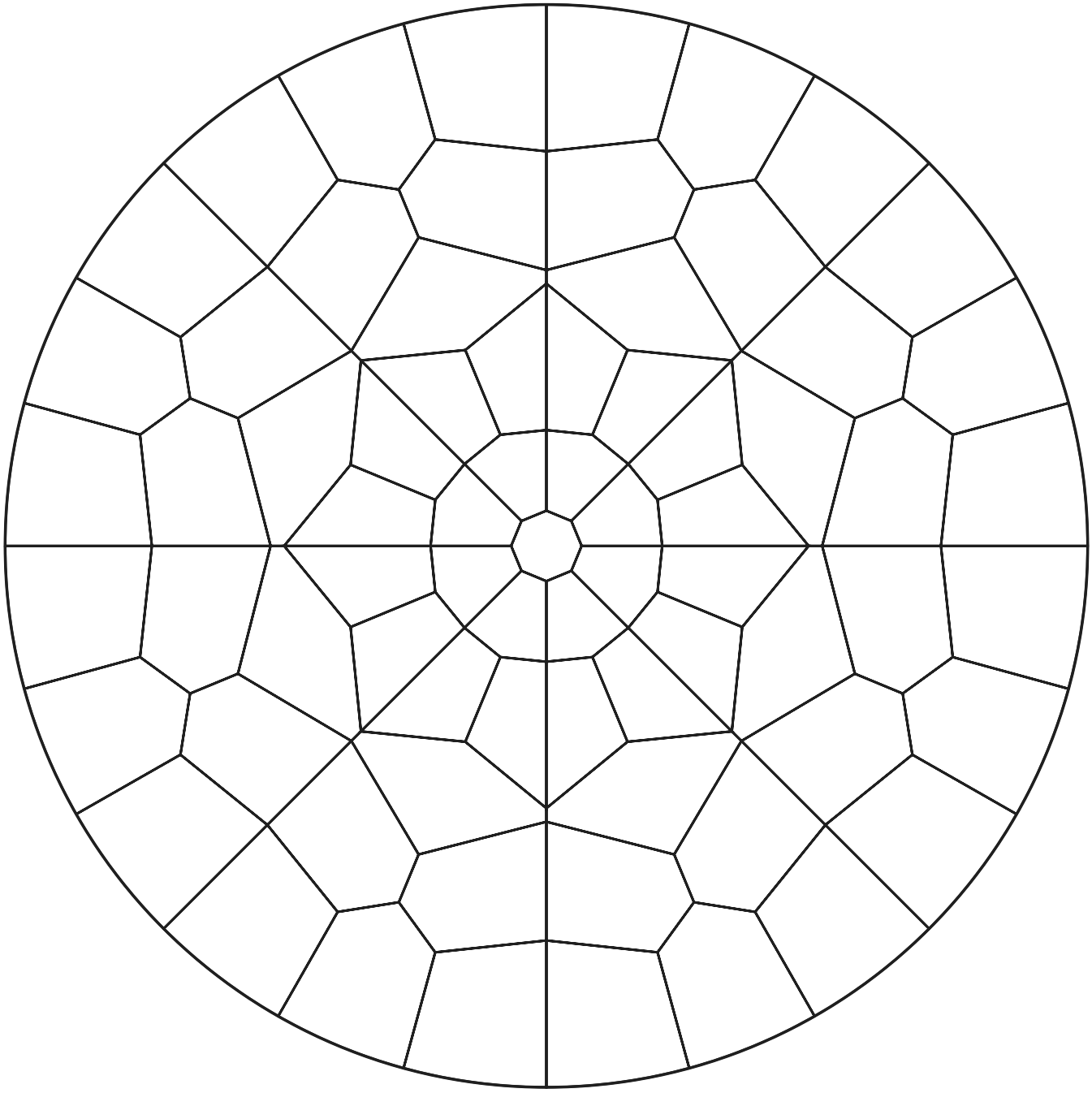
The dual of a Voronoi diagram is the Delaunay triangulation, and the two appear throughout nature and science: in crystal grains, foam, cellular biology, mesh generation and the territories of nesting birds.

Turn the page · fifty gardens of points are waiting to grow.



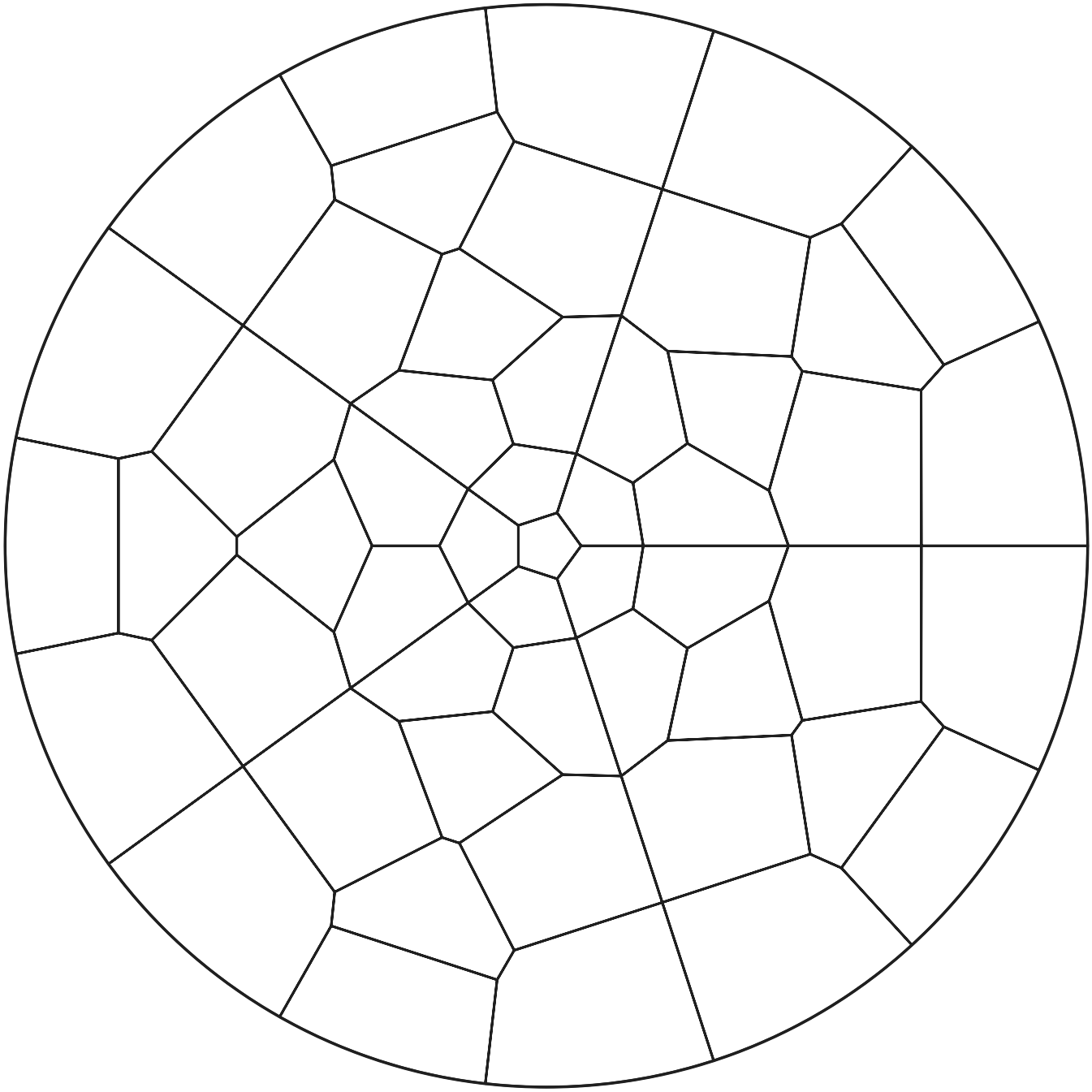
1. Sixfold Bloom

Voronoi Mandala · 6-fold symmetry



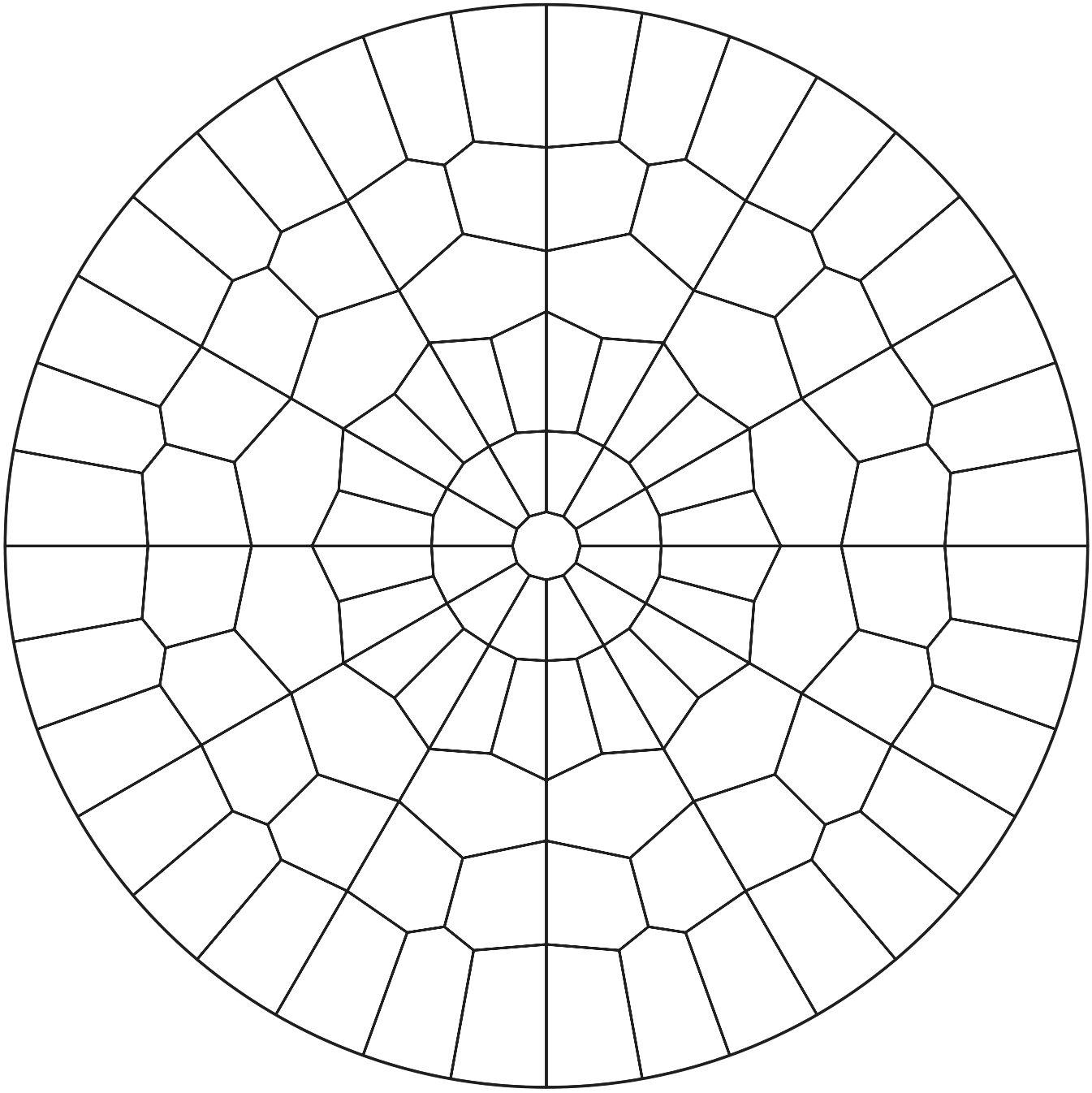
2. Eightfold Rosette

Voronoi Mandala · 8-fold symmetry



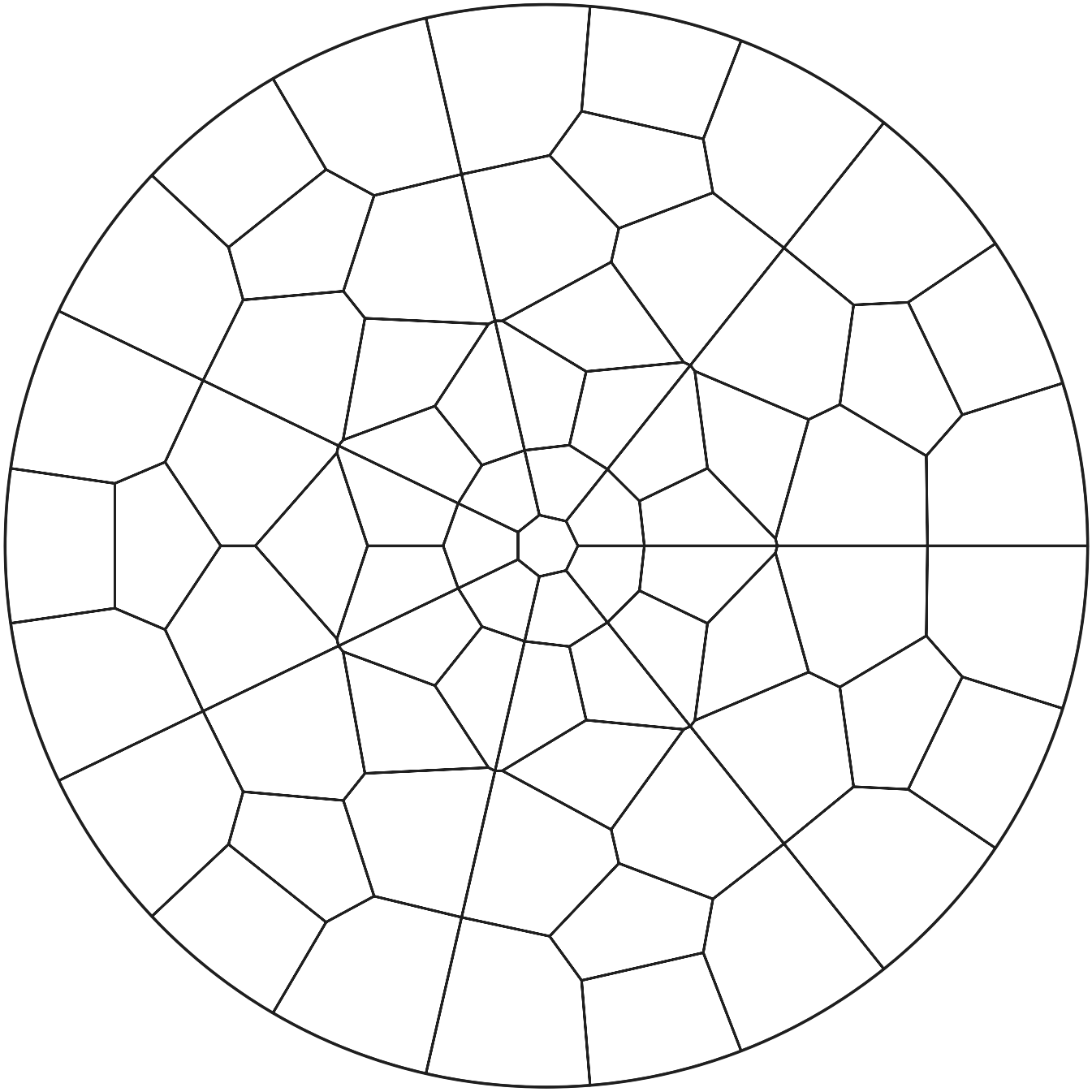
3. Fivefold Star

Voronoi Mandala · 5-fold symmetry



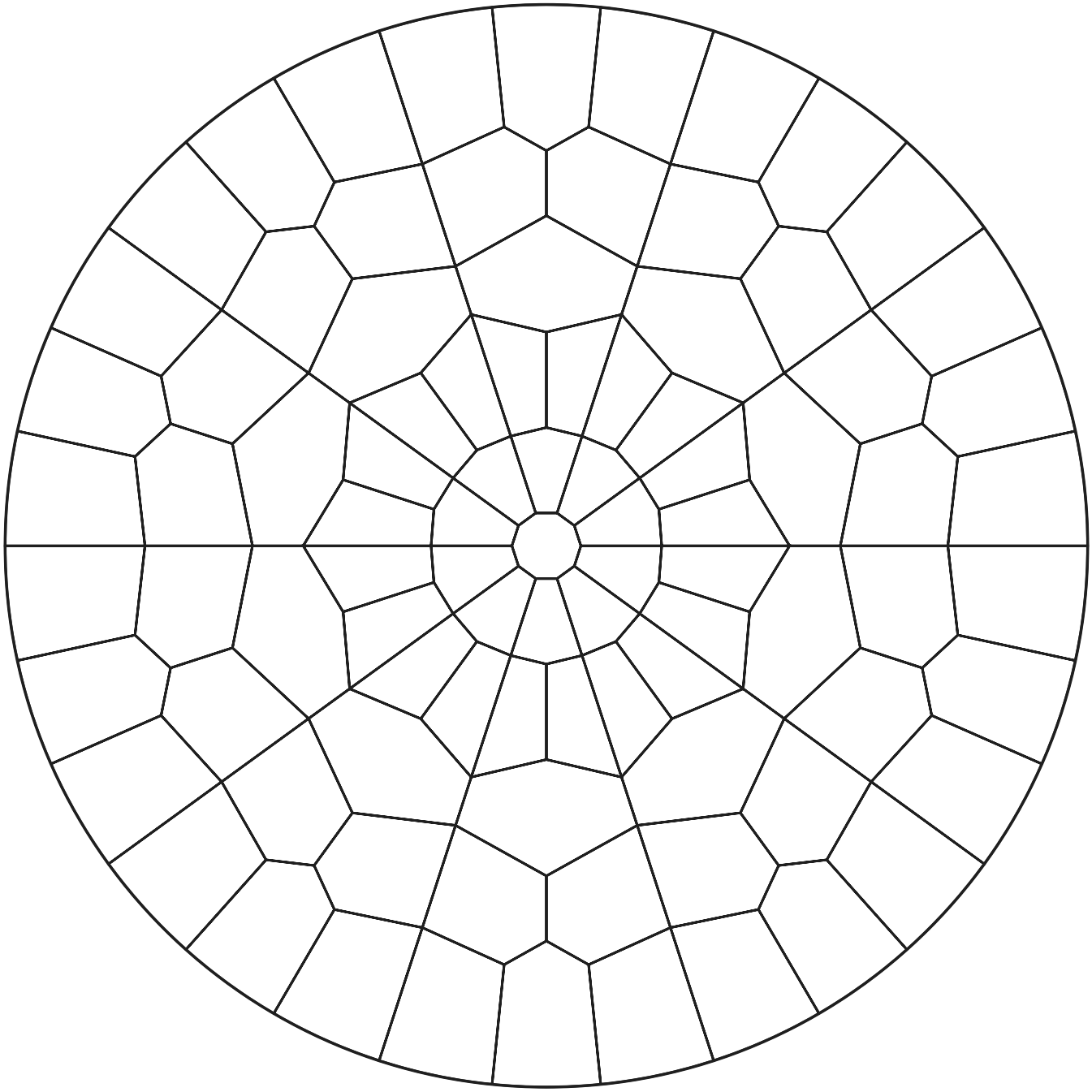
4. Twelvefold Cathedral

Voronoi Mandala · 12-fold symmetry



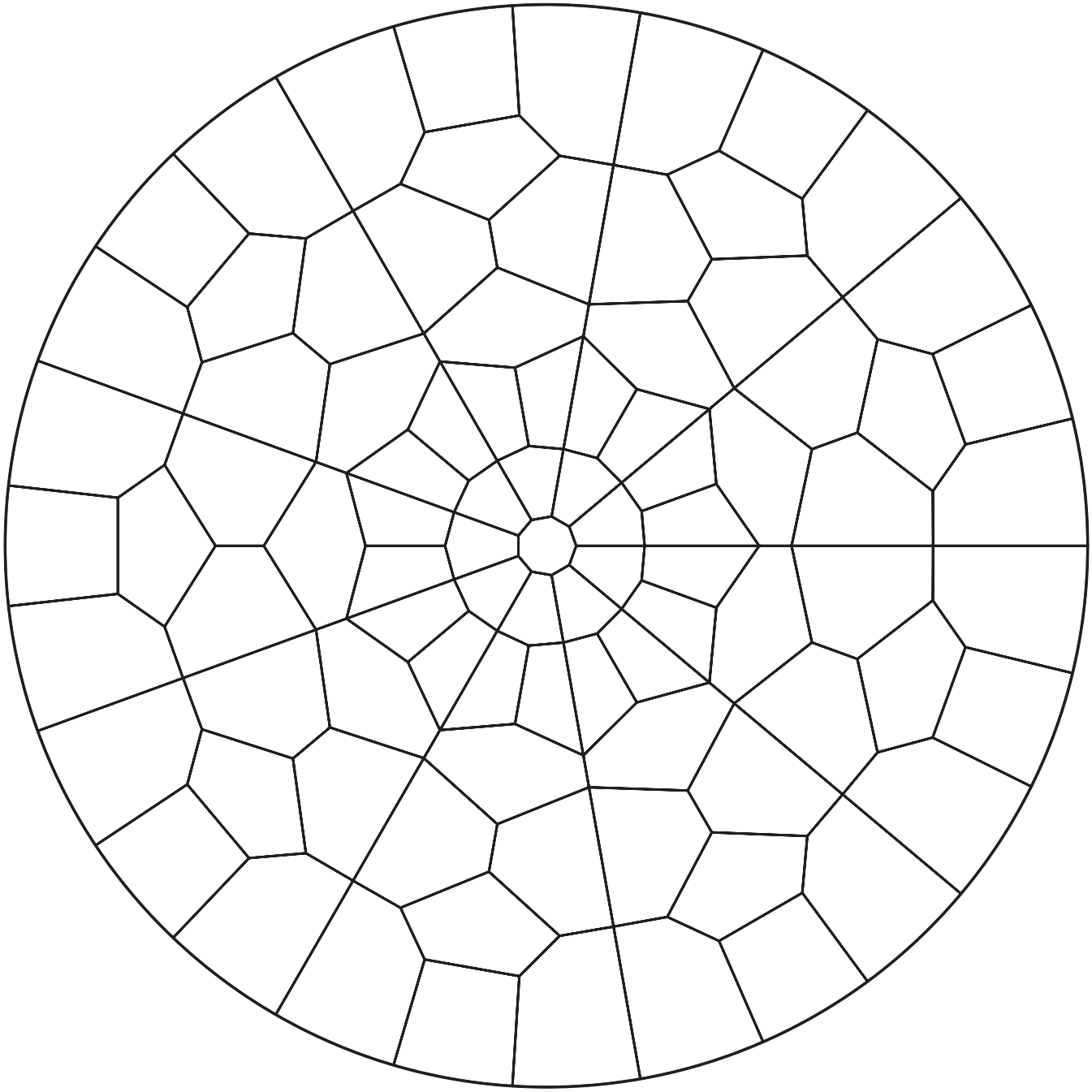
5. Sevenfold Lotus

Voronoi Mandala · 7-fold symmetry



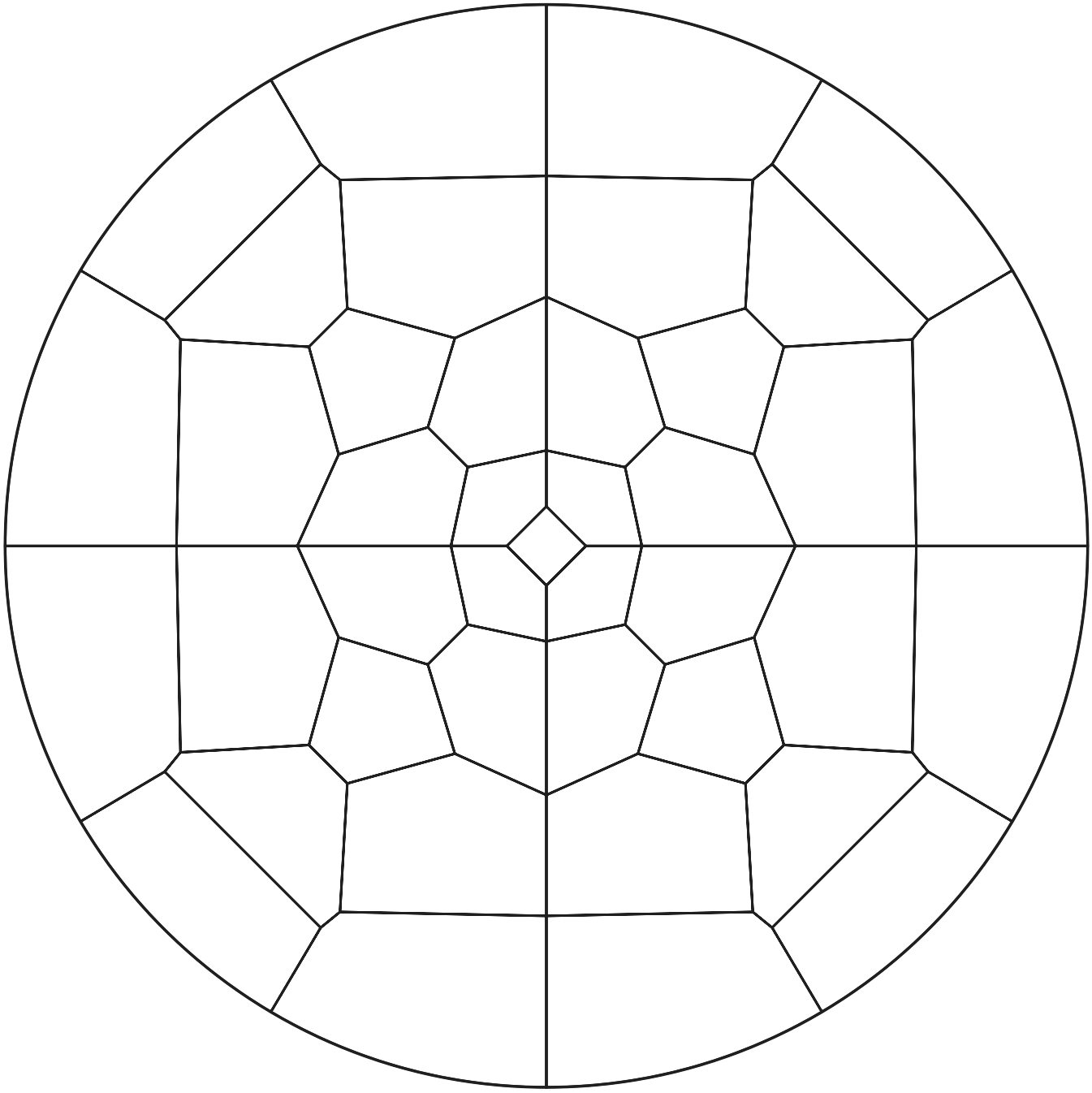
6. Tenfold Compass

Voronoi Mandala · 10-fold symmetry



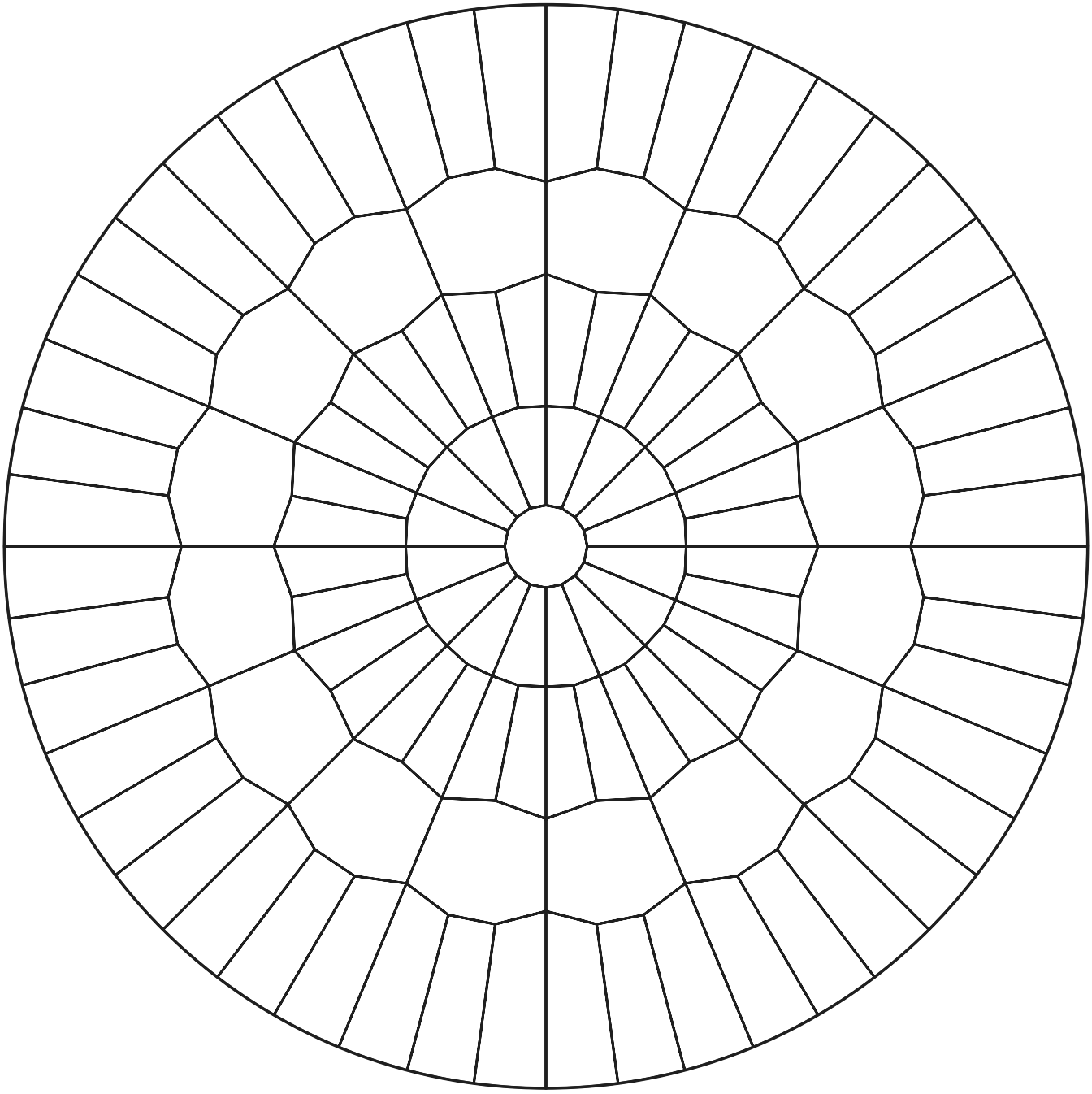
7. Ninefold Wheel

Voronoi Mandala · 9-fold symmetry



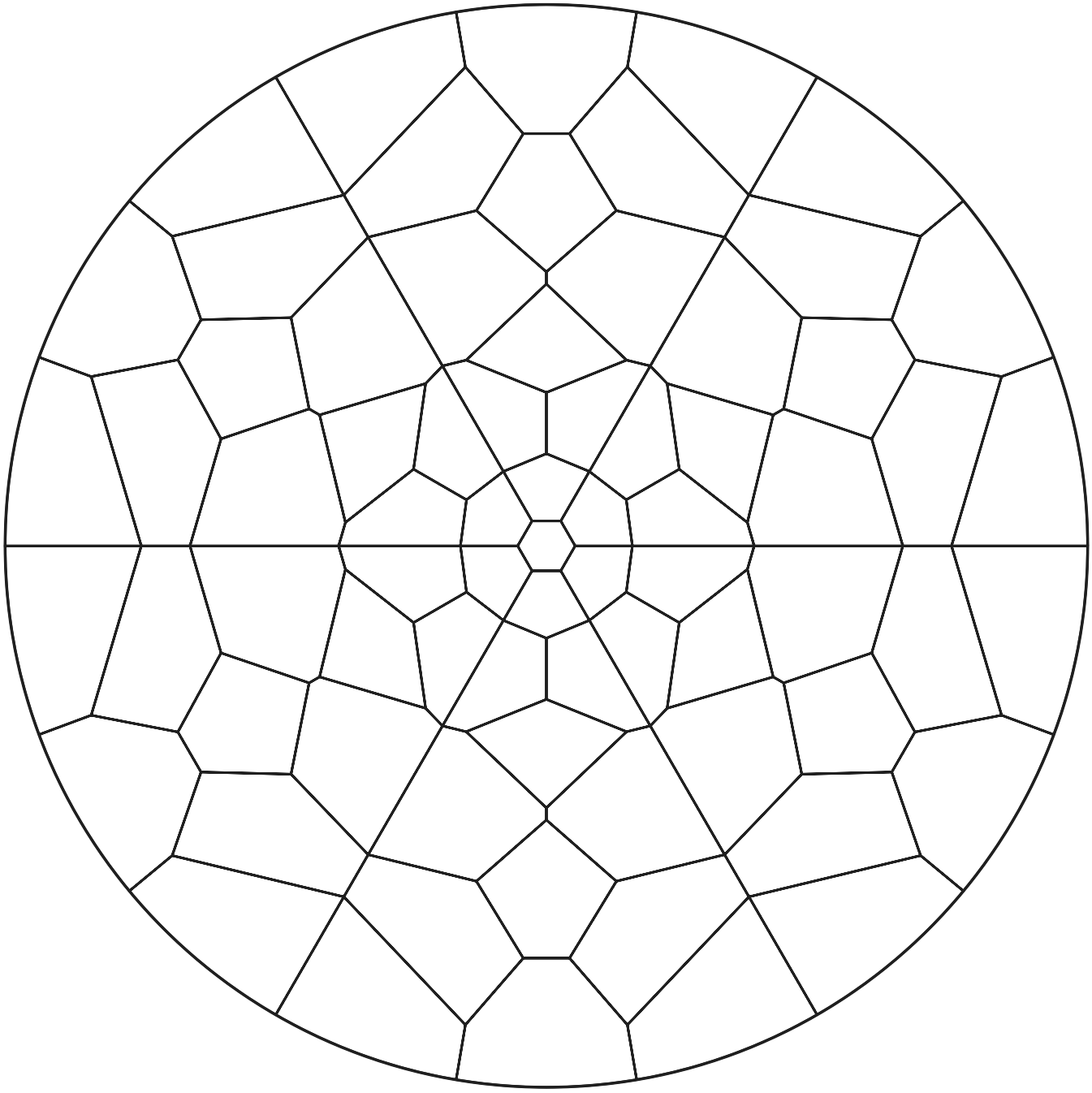
8. Fourfold Crossing

Voronoi Mandala · 4-fold symmetry



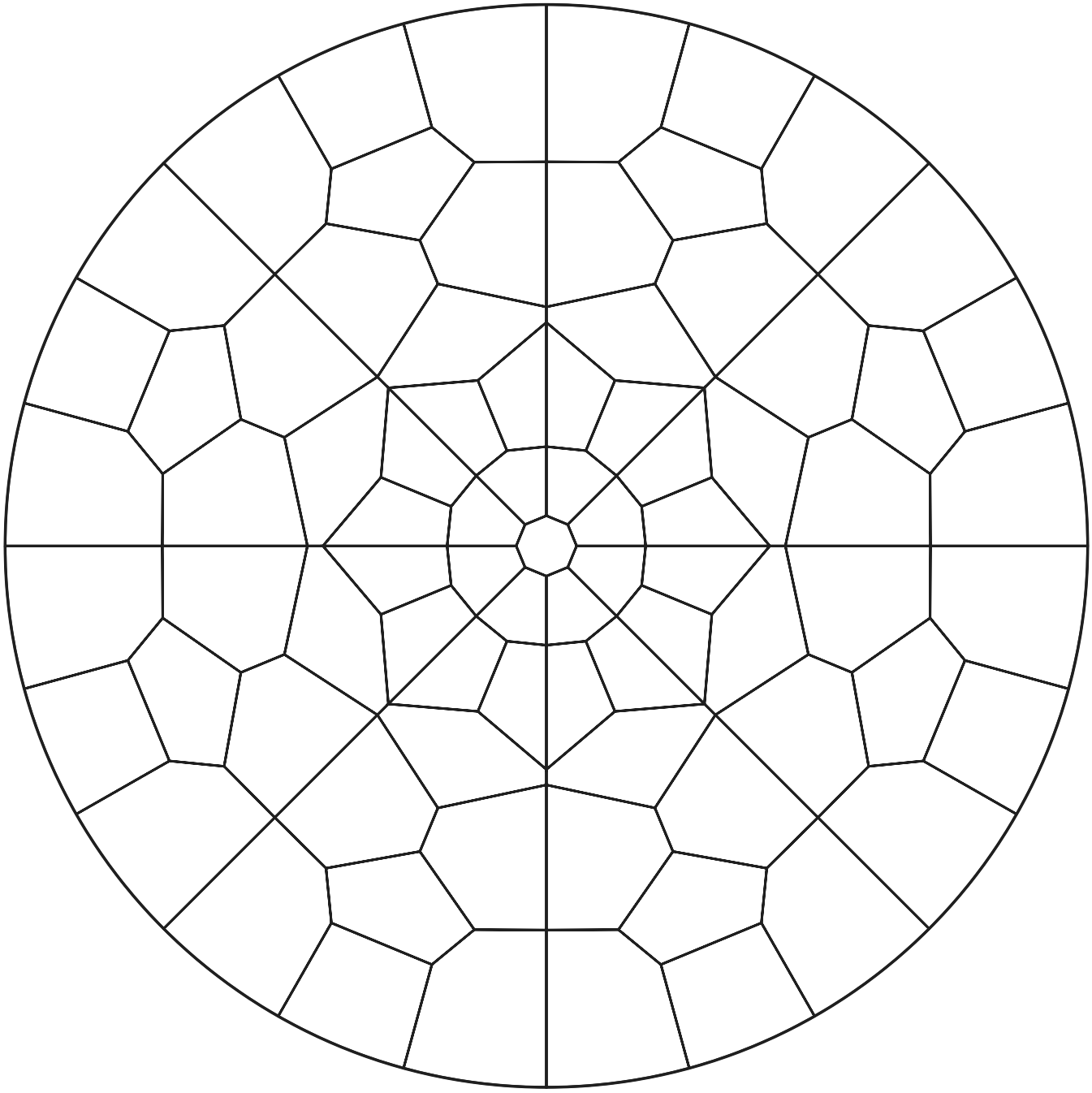
9. Sixteenfold Halo

Voronoi Mandala · 16-fold symmetry



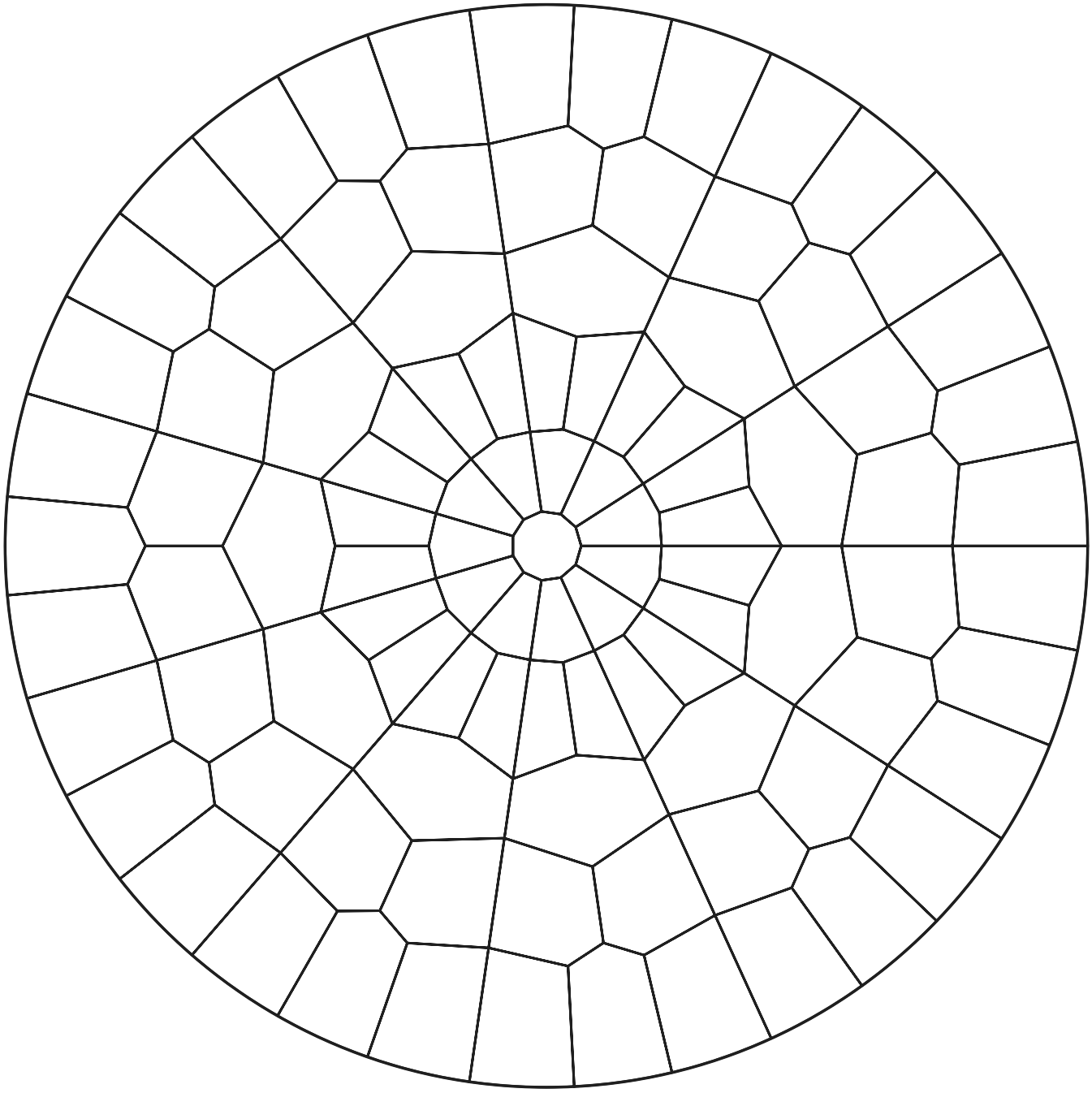
10. Sixfold Snowflake

Voronoi Mandala · 6-fold symmetry



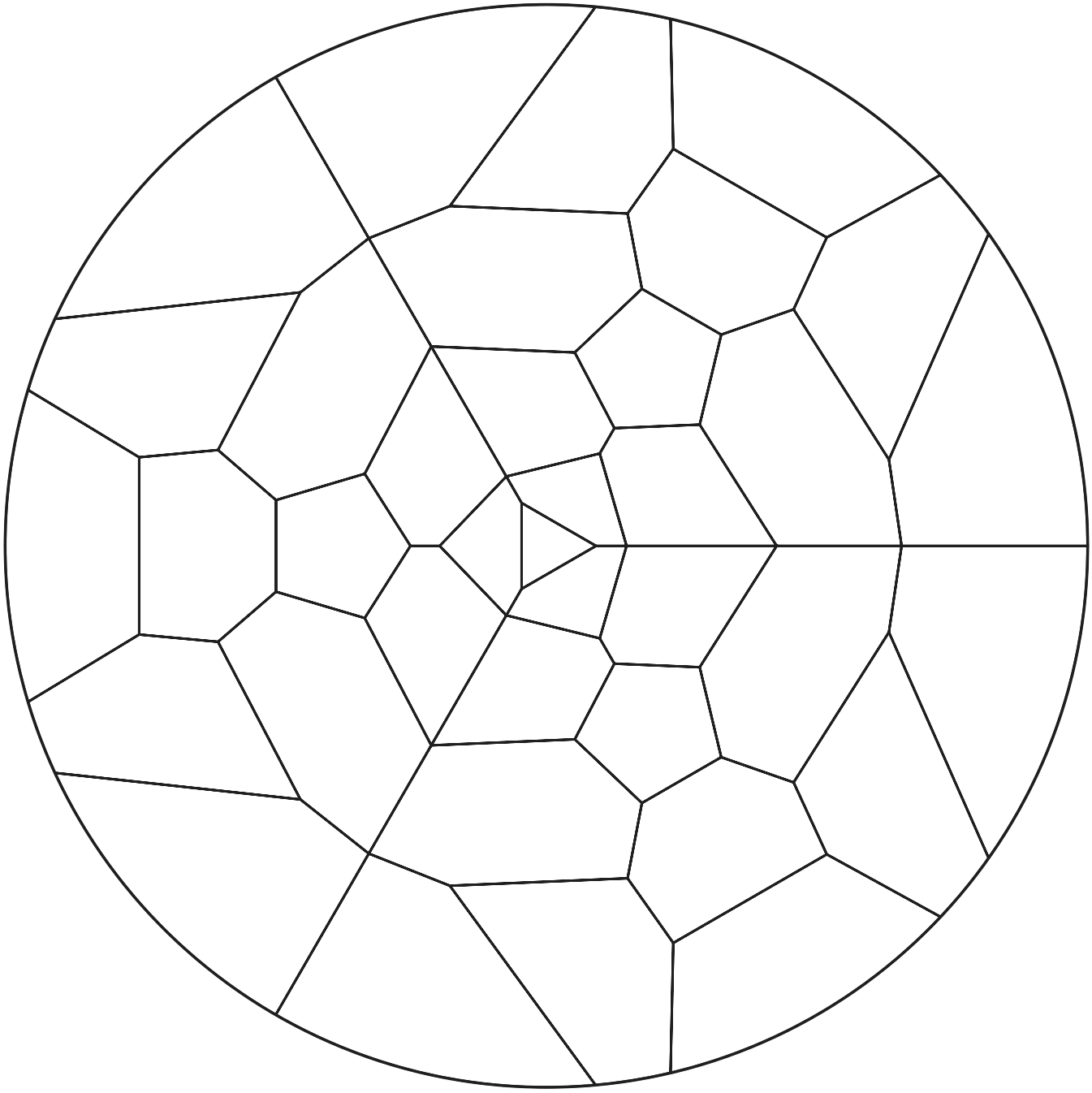
11. Eightfold Mariner

Voronoi Mandala · 8-fold symmetry



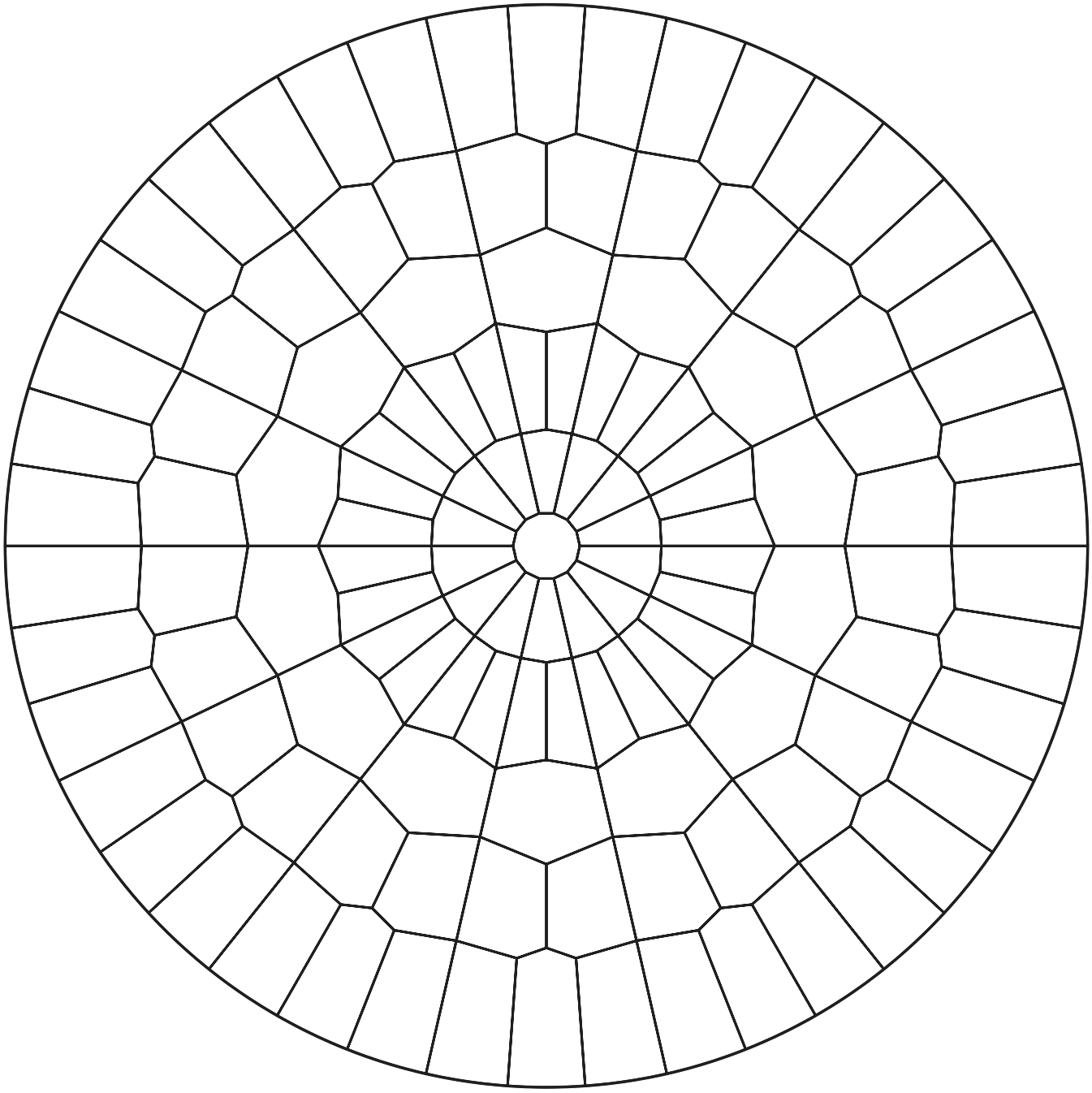
12. Elevenfold Crown

Voronoi Mandala · 11-fold symmetry



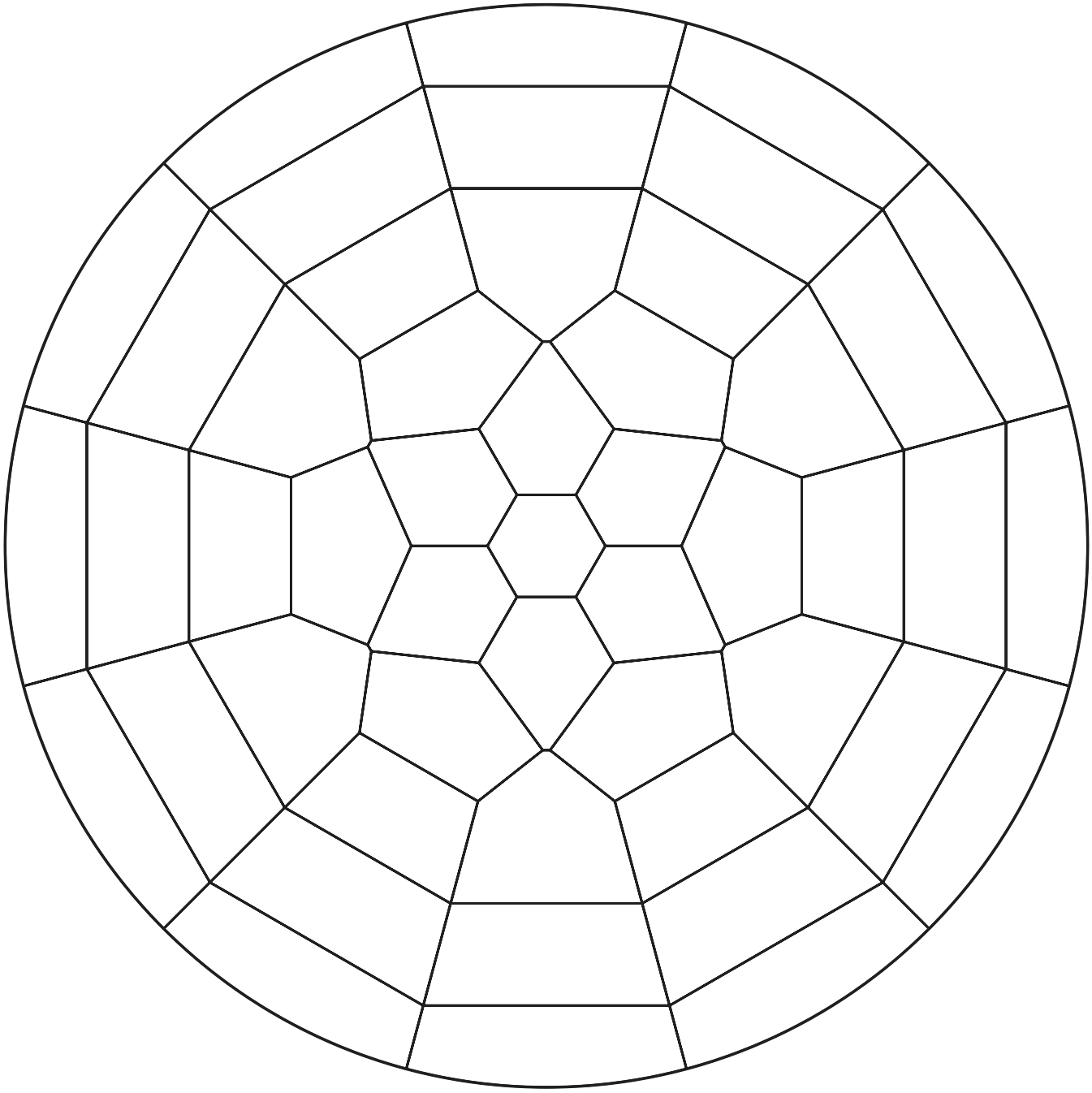
13. Threefold Trefoil

Voronoi Mandala · 3-fold symmetry



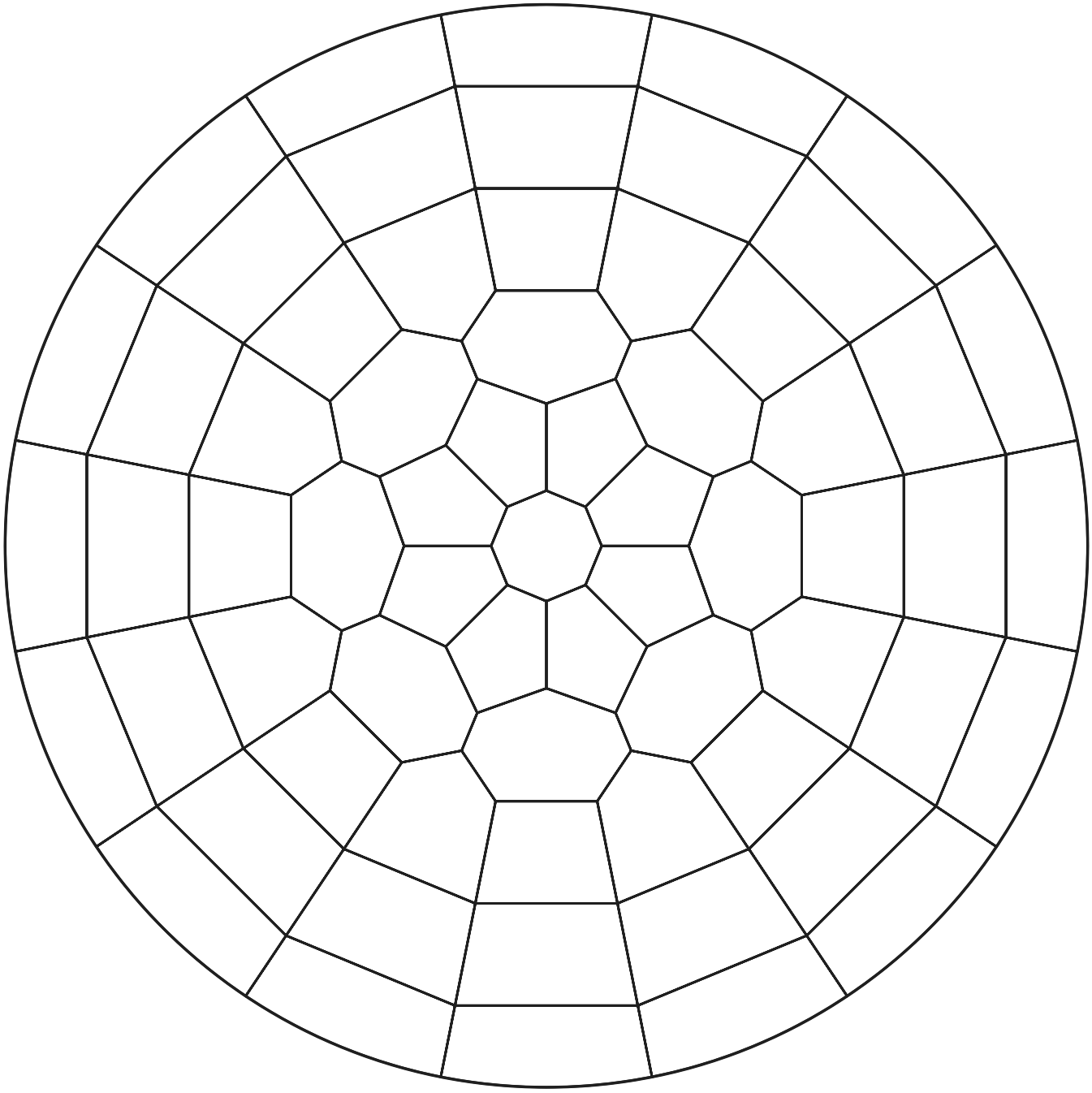
14. Fourteenfold Sun

Voronoi Mandala · 14-fold symmetry



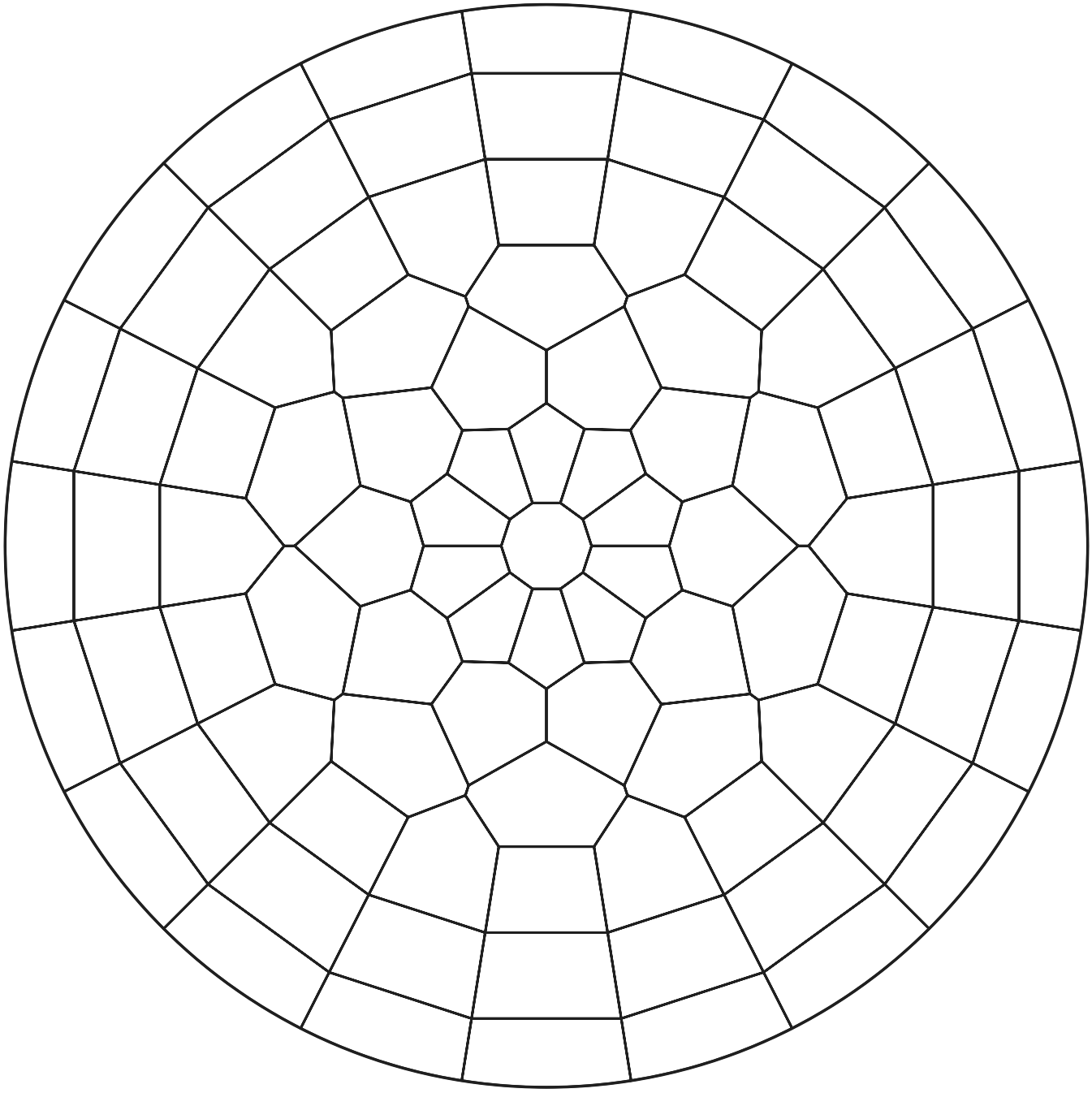
15. Rose Window I

Concentric Voronoi · radial rings



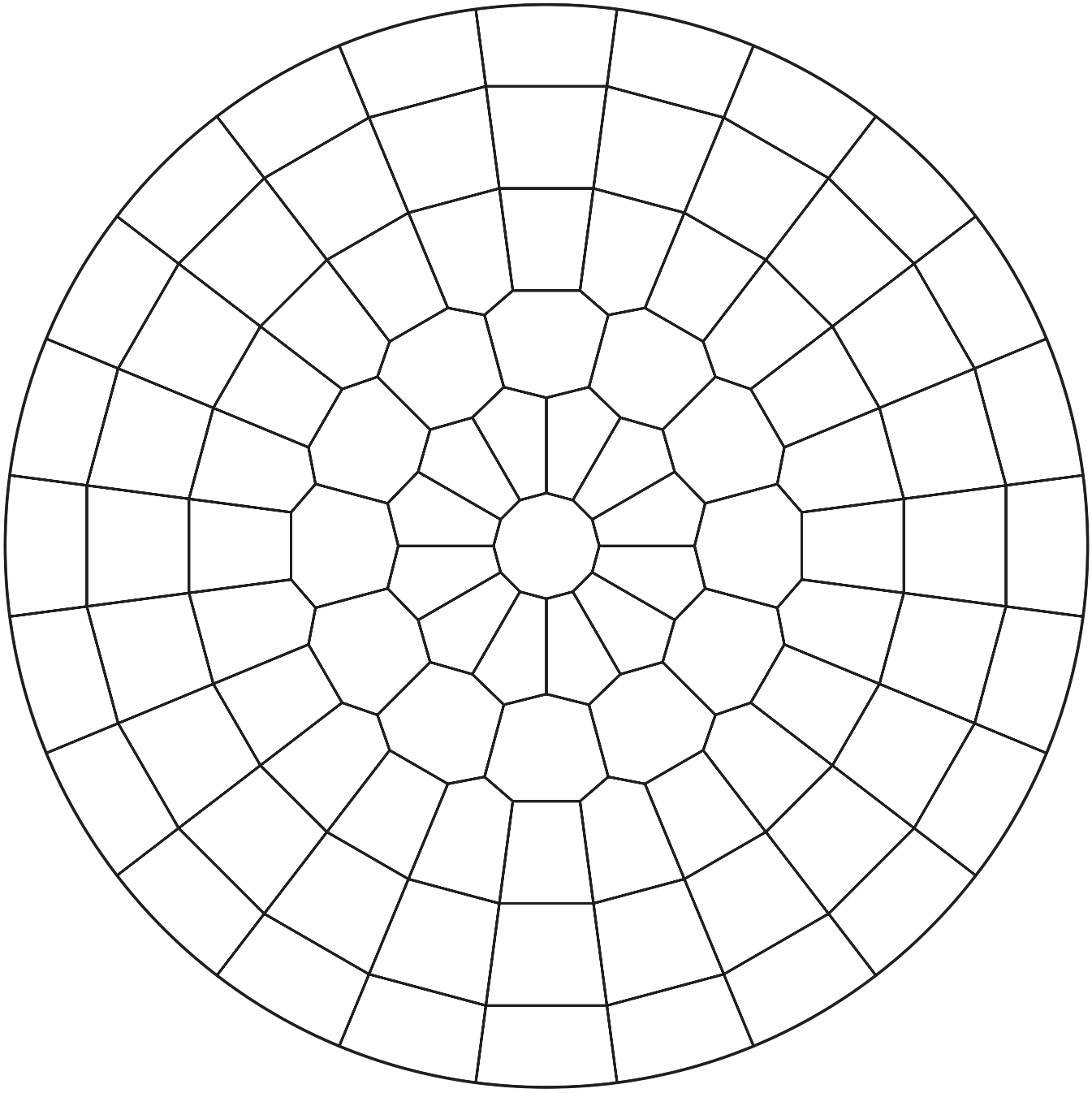
16. Rose Window II

Concentric Voronoi · radial rings



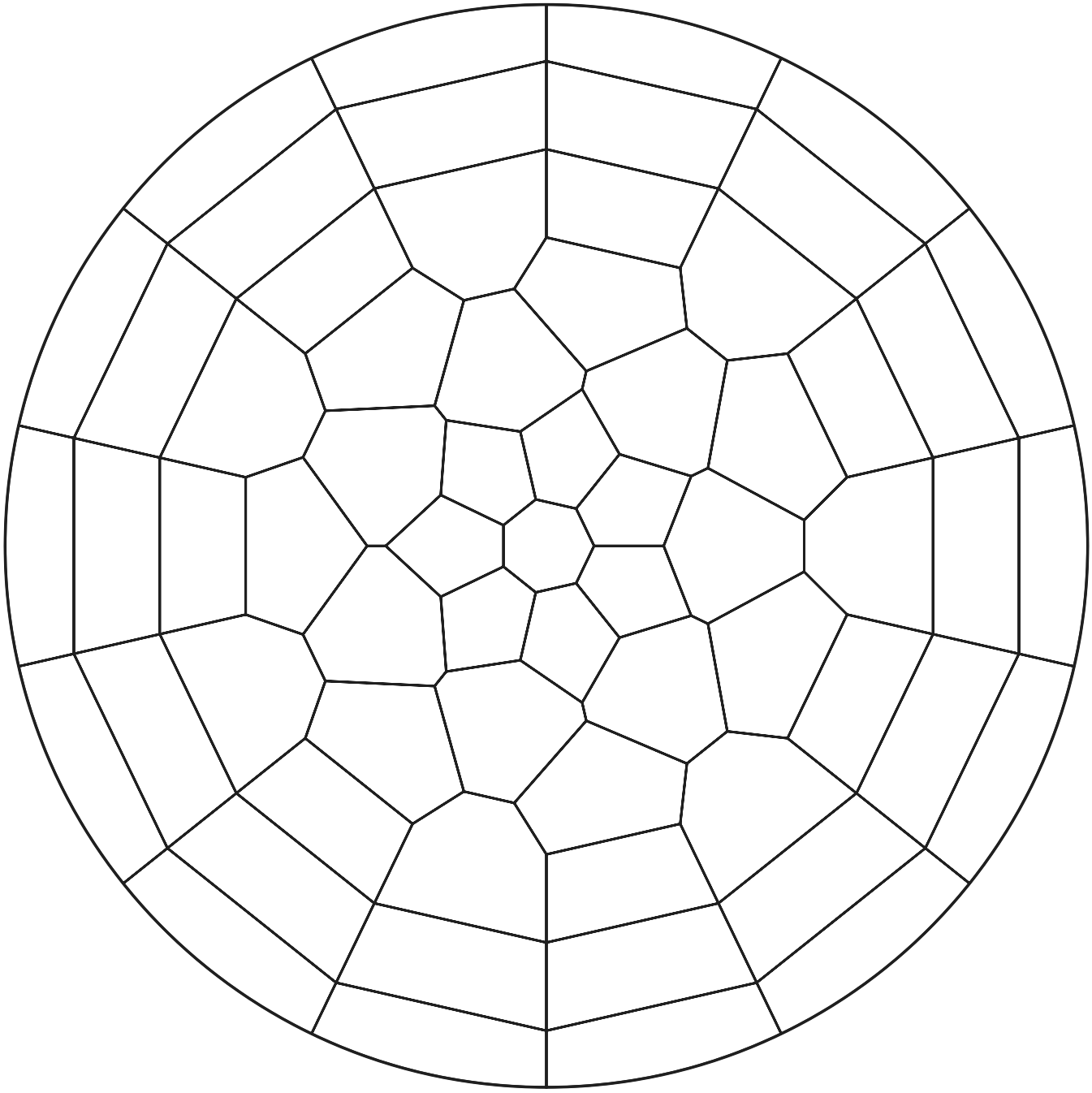
17. Rose Window III

Concentric Voronoi · radial rings



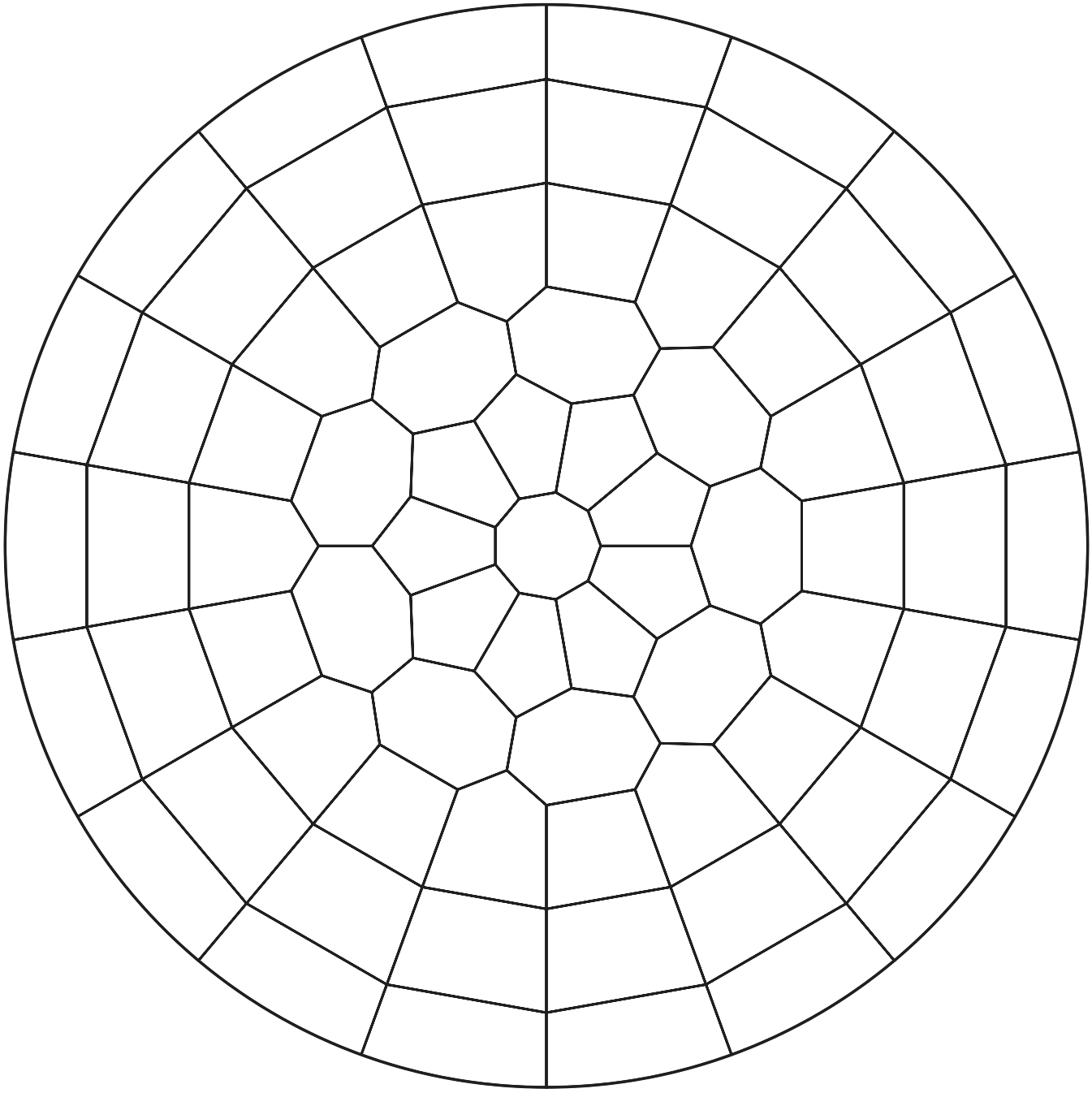
18. Gothic Oculus

Concentric Voronoi · radial rings



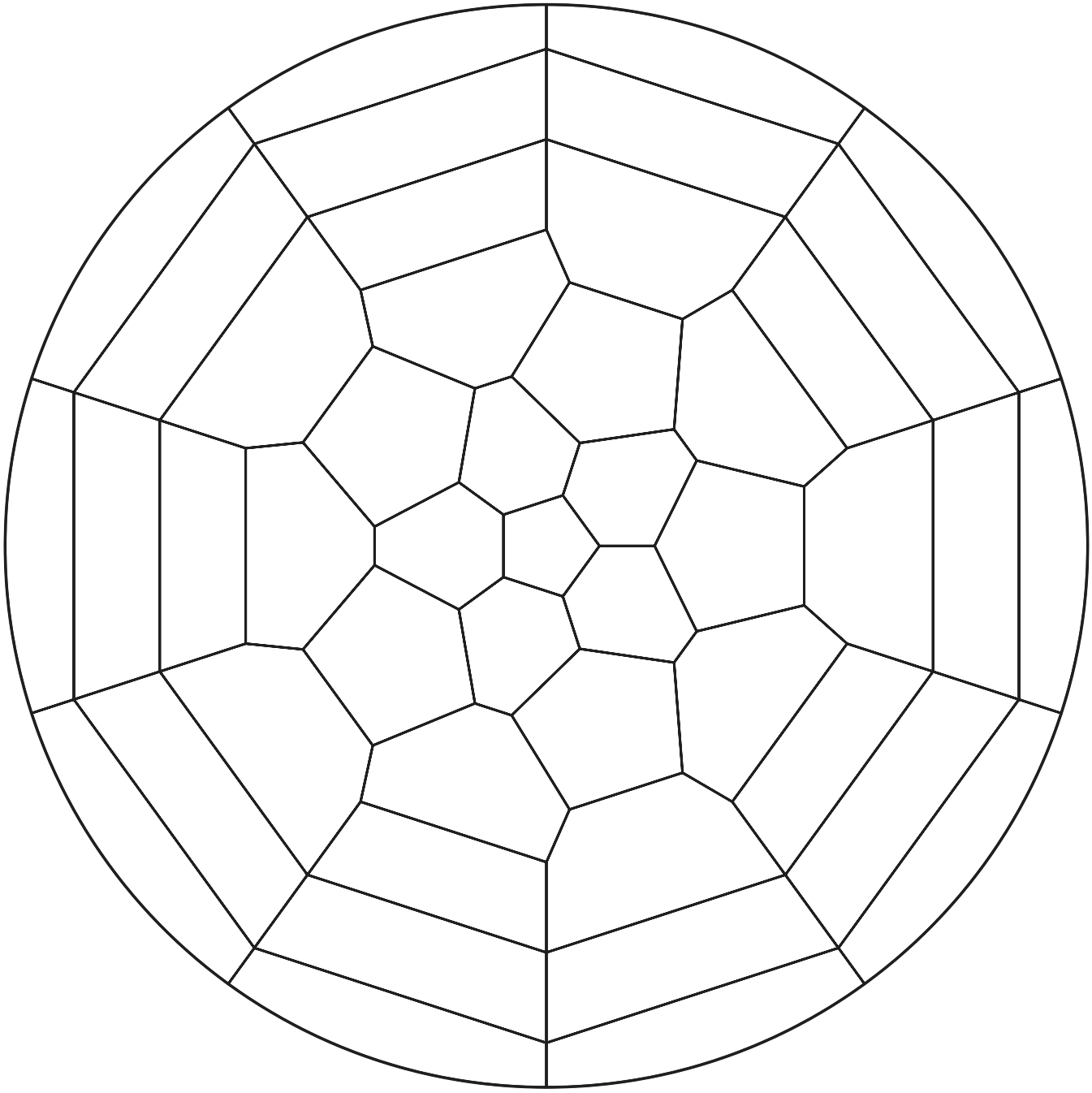
19. Solar Disc

Concentric Voronoi · radial rings



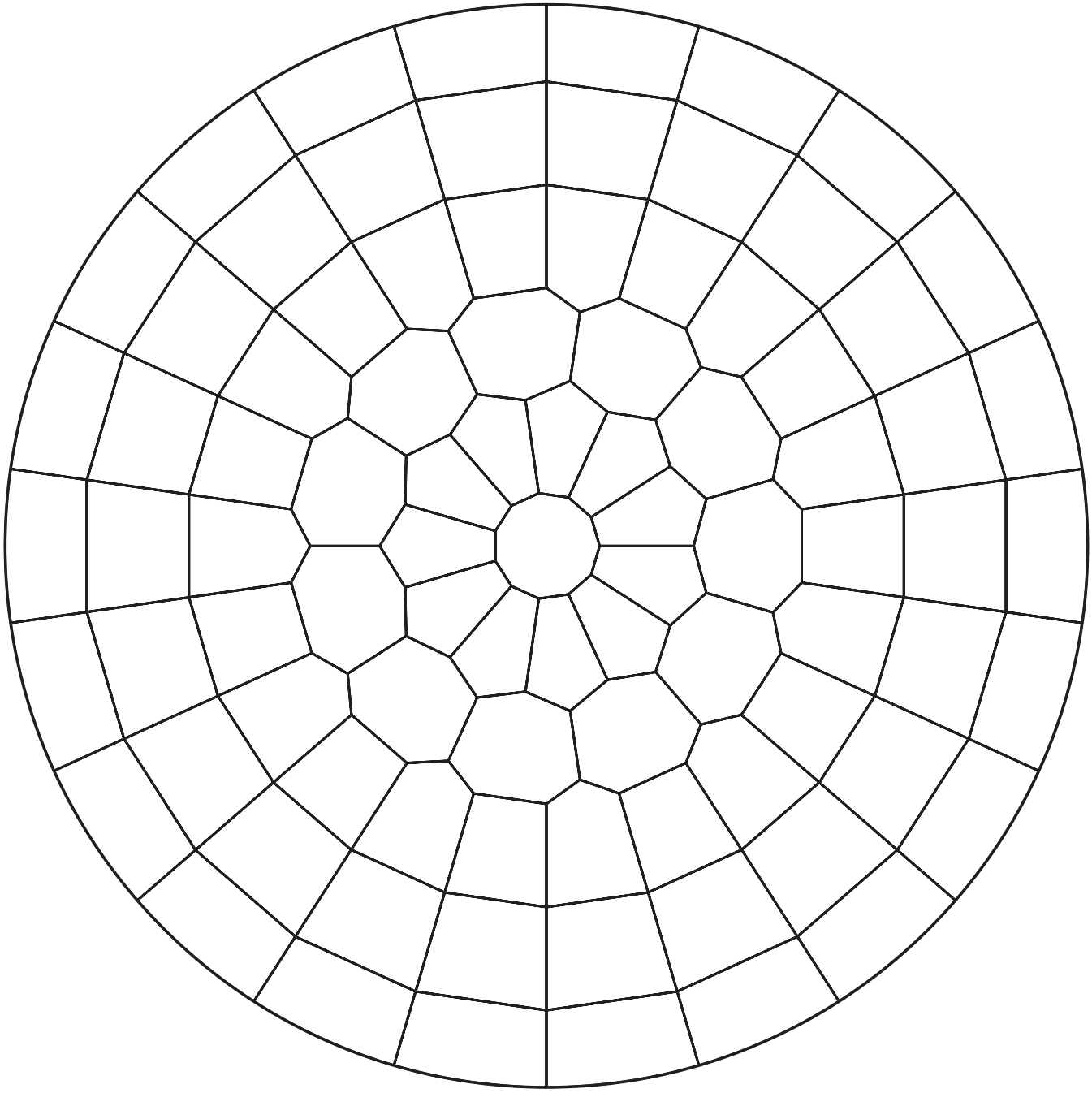
20. Mosaic Wheel

Concentric Voronoi · radial rings



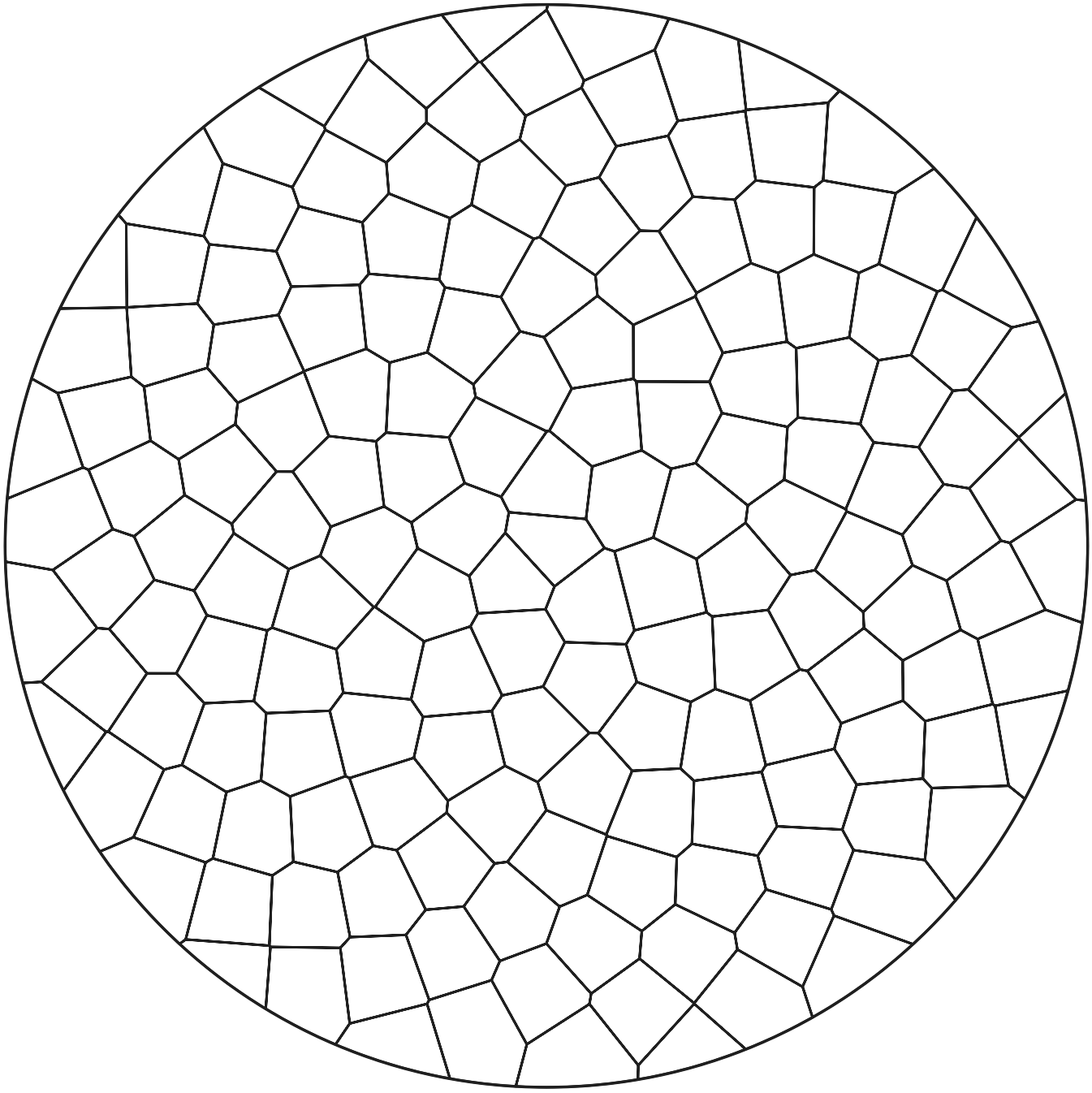
21. Lunar Ring

Concentric Voronoi · radial rings



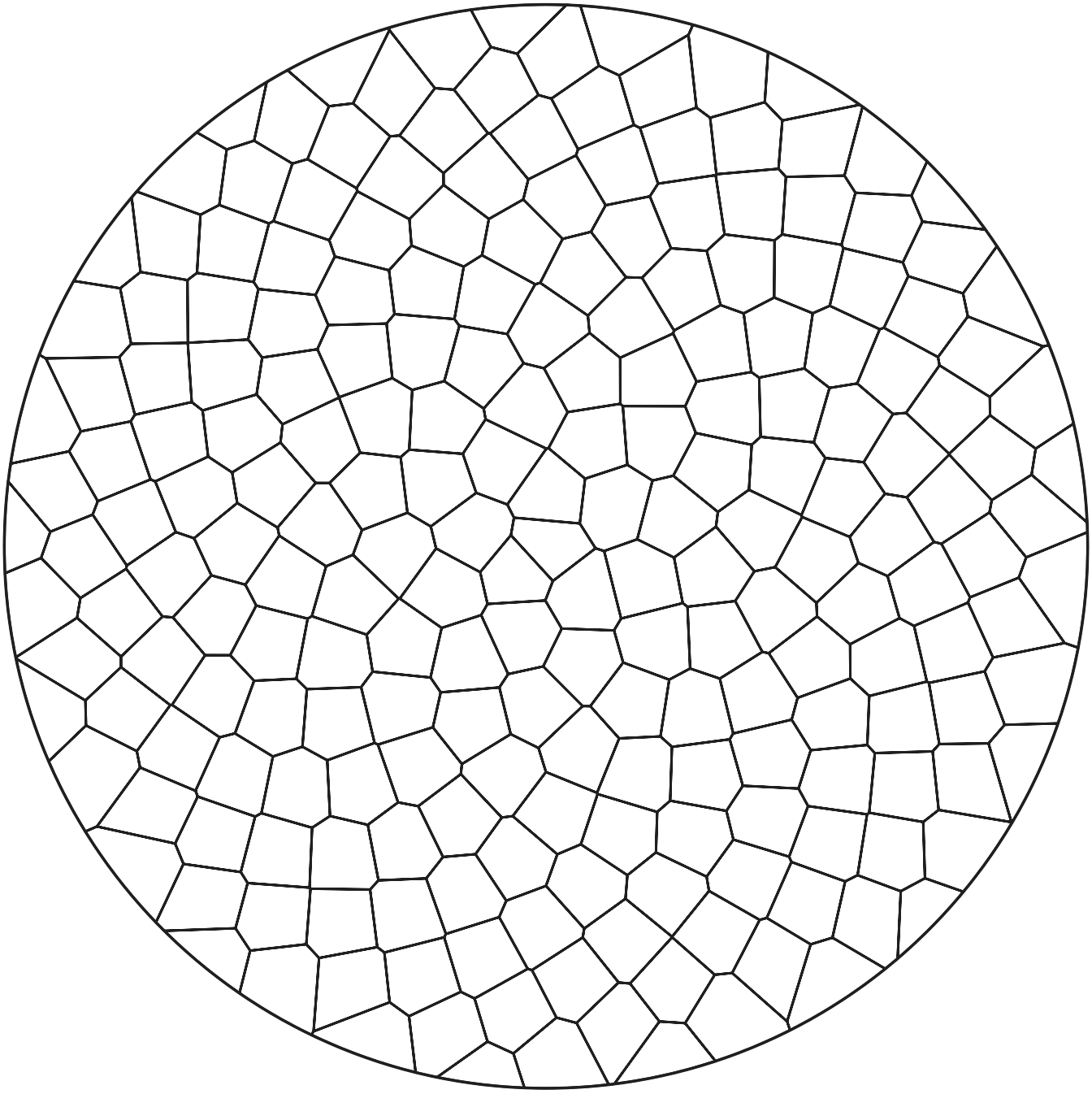
22. Clockwork Mandala

Concentric Voronoi · radial rings



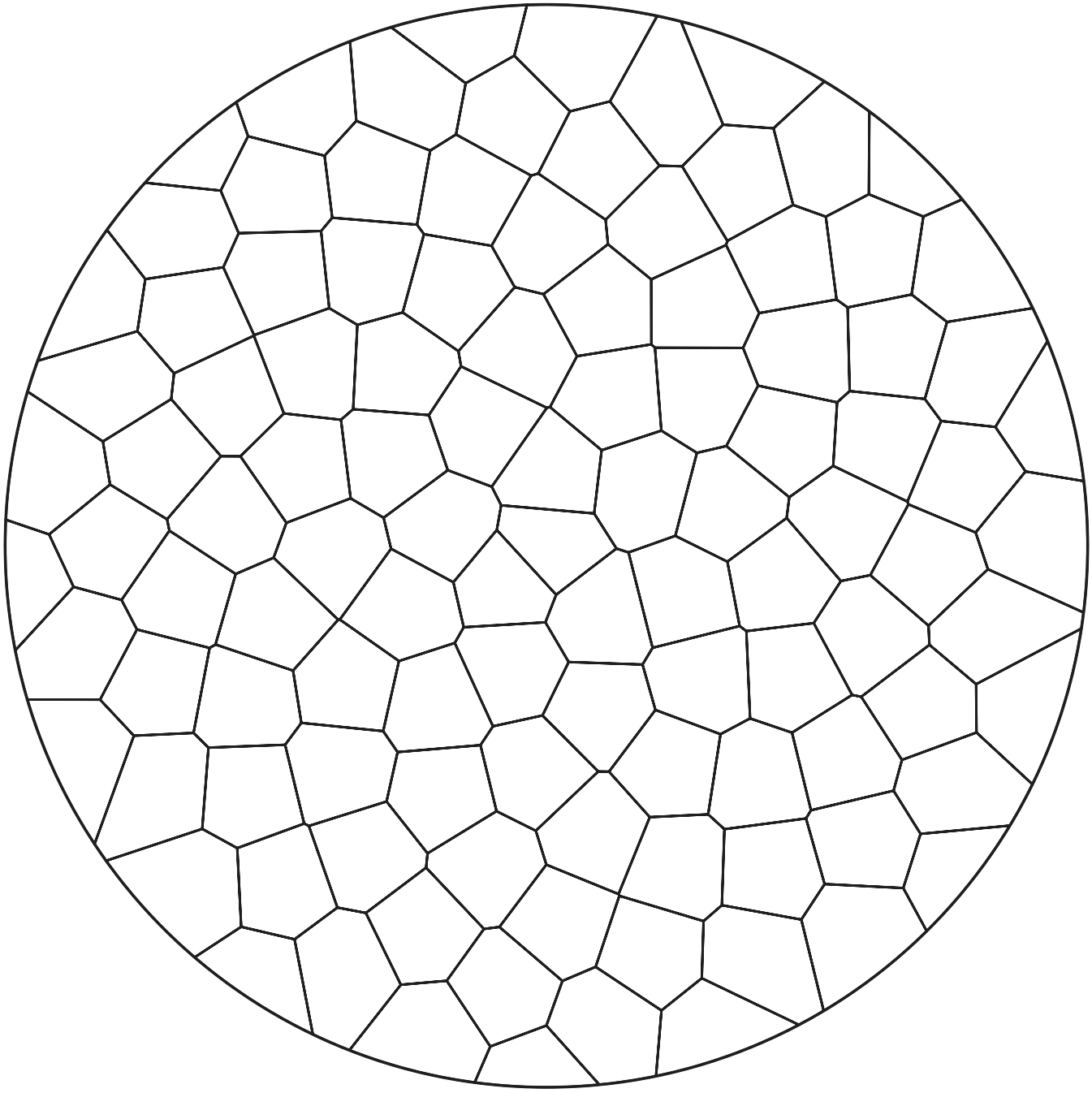
23. Sunflower Lattice

Phyllotaxis Voronoi · golden angle 137.5°



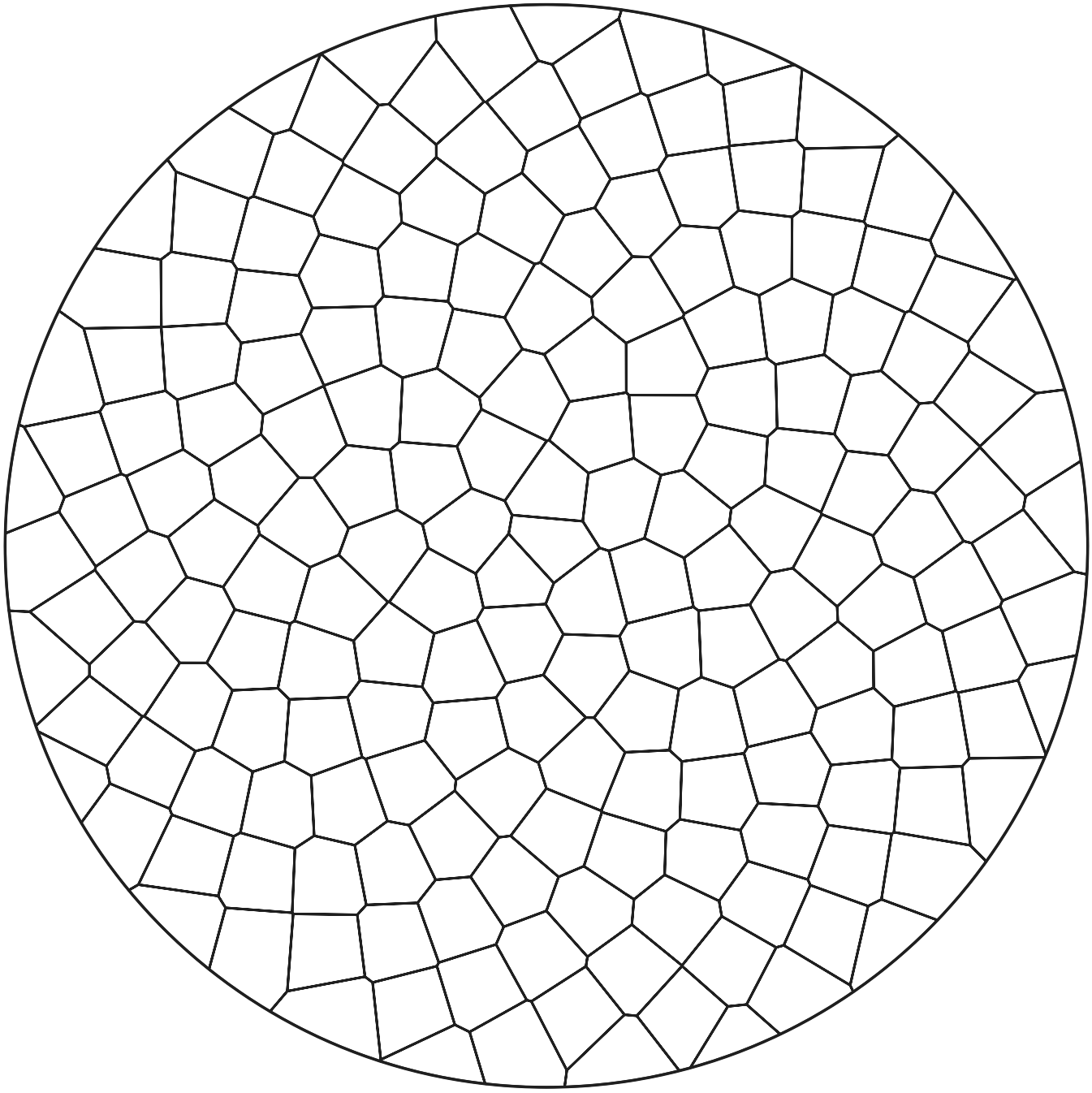
24. Golden Spiral Field

Phyllotaxis Voronoi · golden angle 137.5°



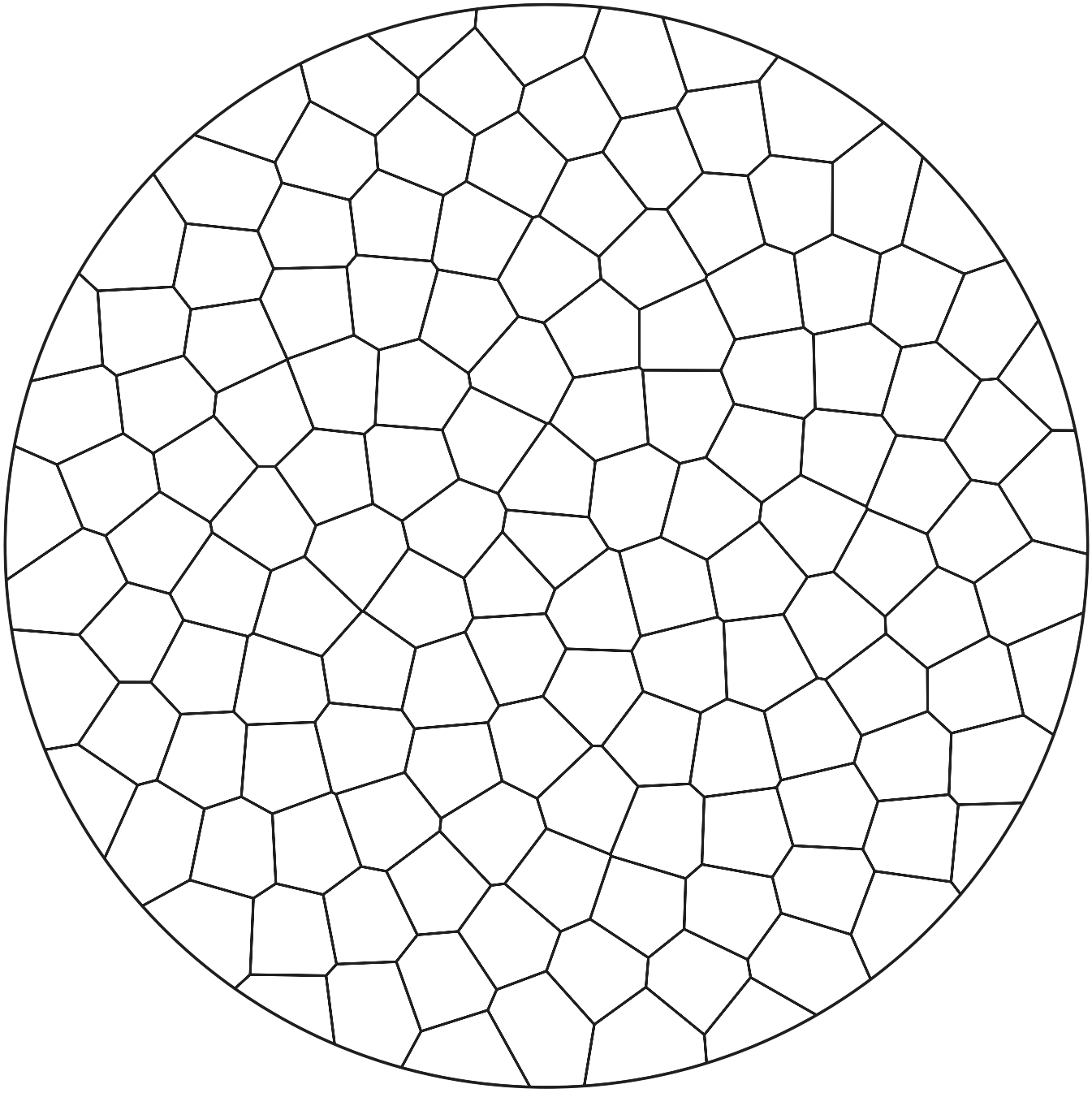
25. Seed Head

Phyllotaxis Voronoi · golden angle 137.5°



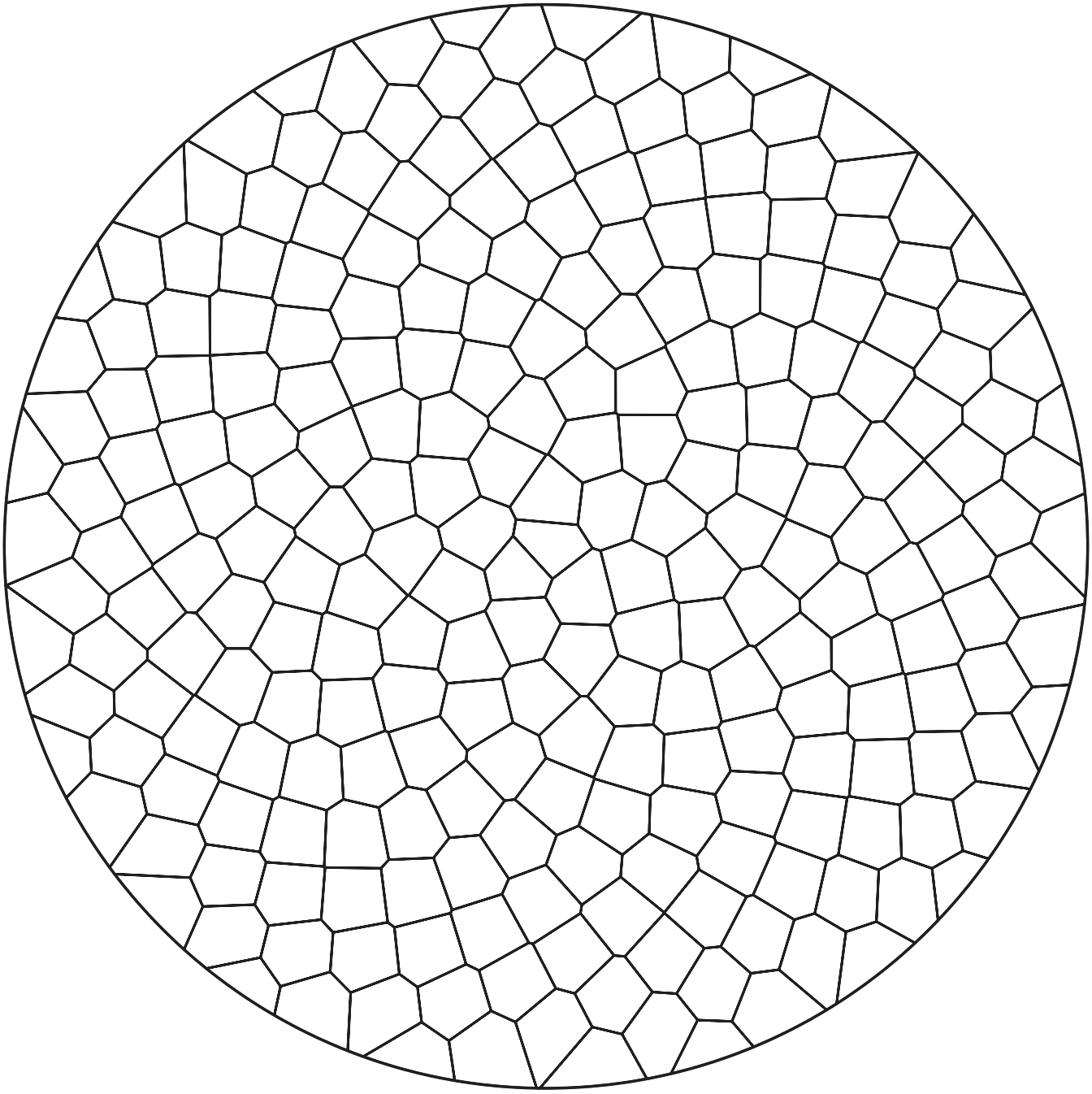
26. Daisy Heart

Phyllotaxis Voronoi · golden angle 137.5°



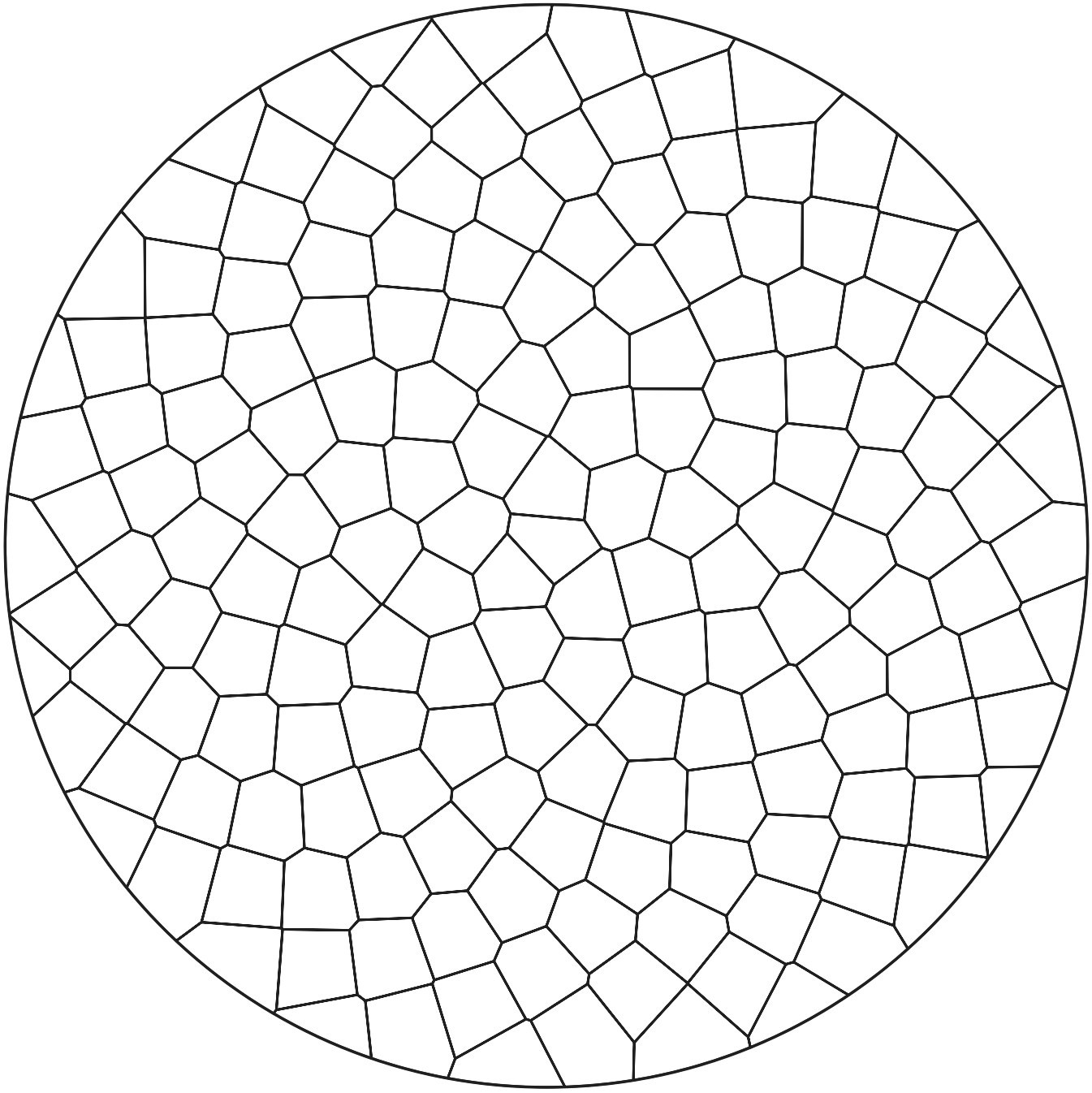
27. Pinecone Spiral

Phyllotaxis Voronoi · golden angle 137.5°



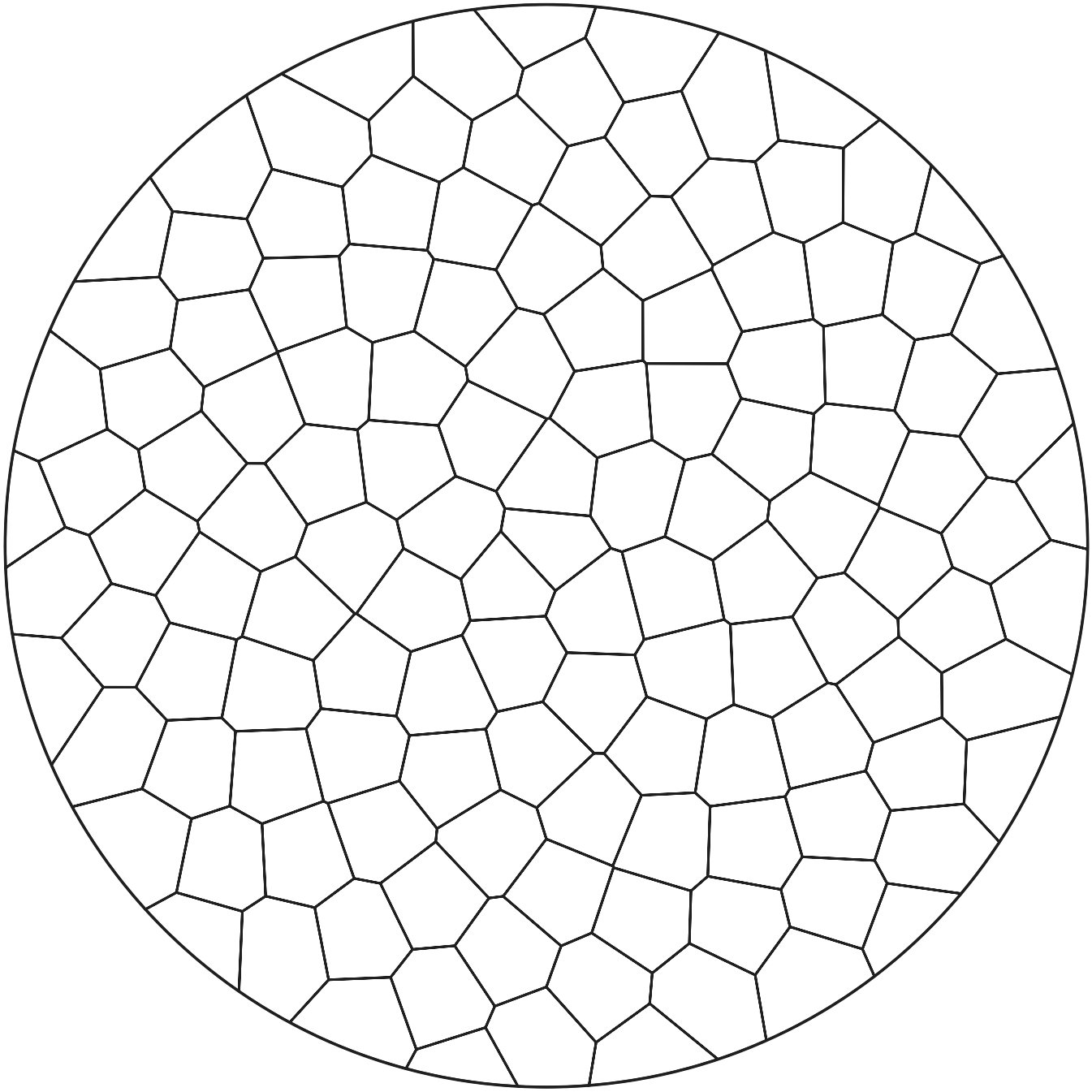
28. Fibonacci Bloom

Phyllotaxis Voronoi · golden angle 137.5°



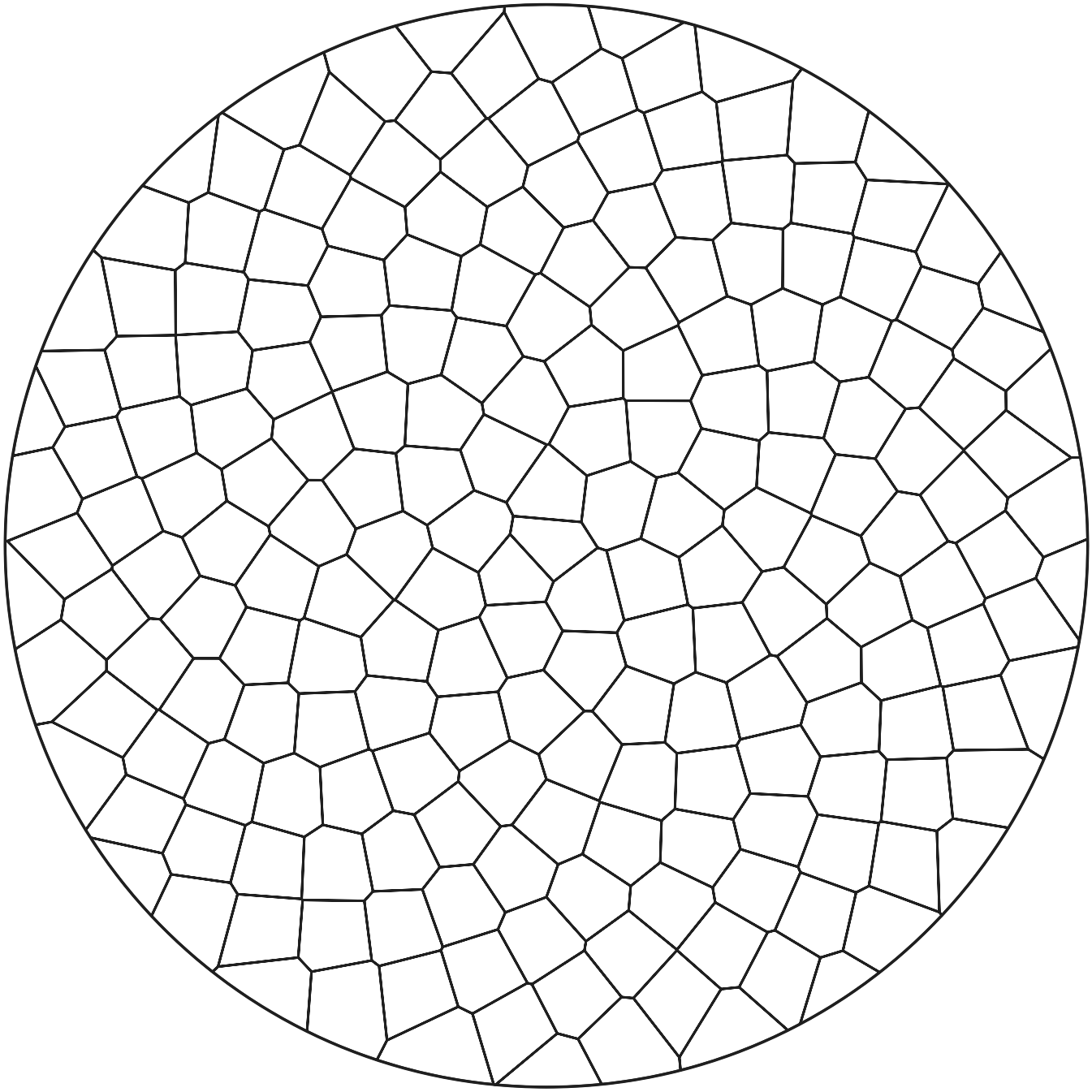
29. Aster Field

Phyllotaxis Voronoi · golden angle 137.5°



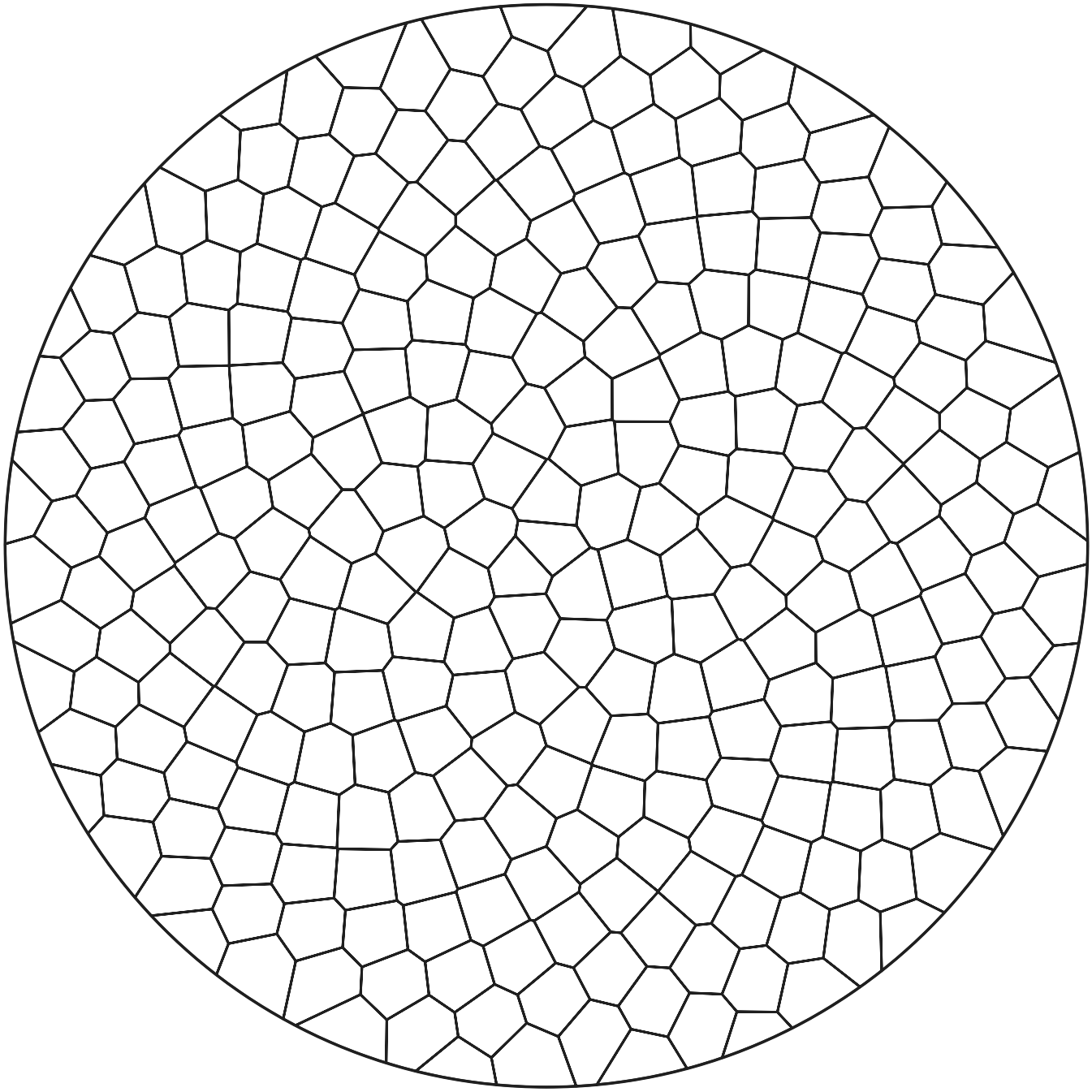
30. Dandelion Clock

Phyllotaxis Voronoi · golden angle 137.5°



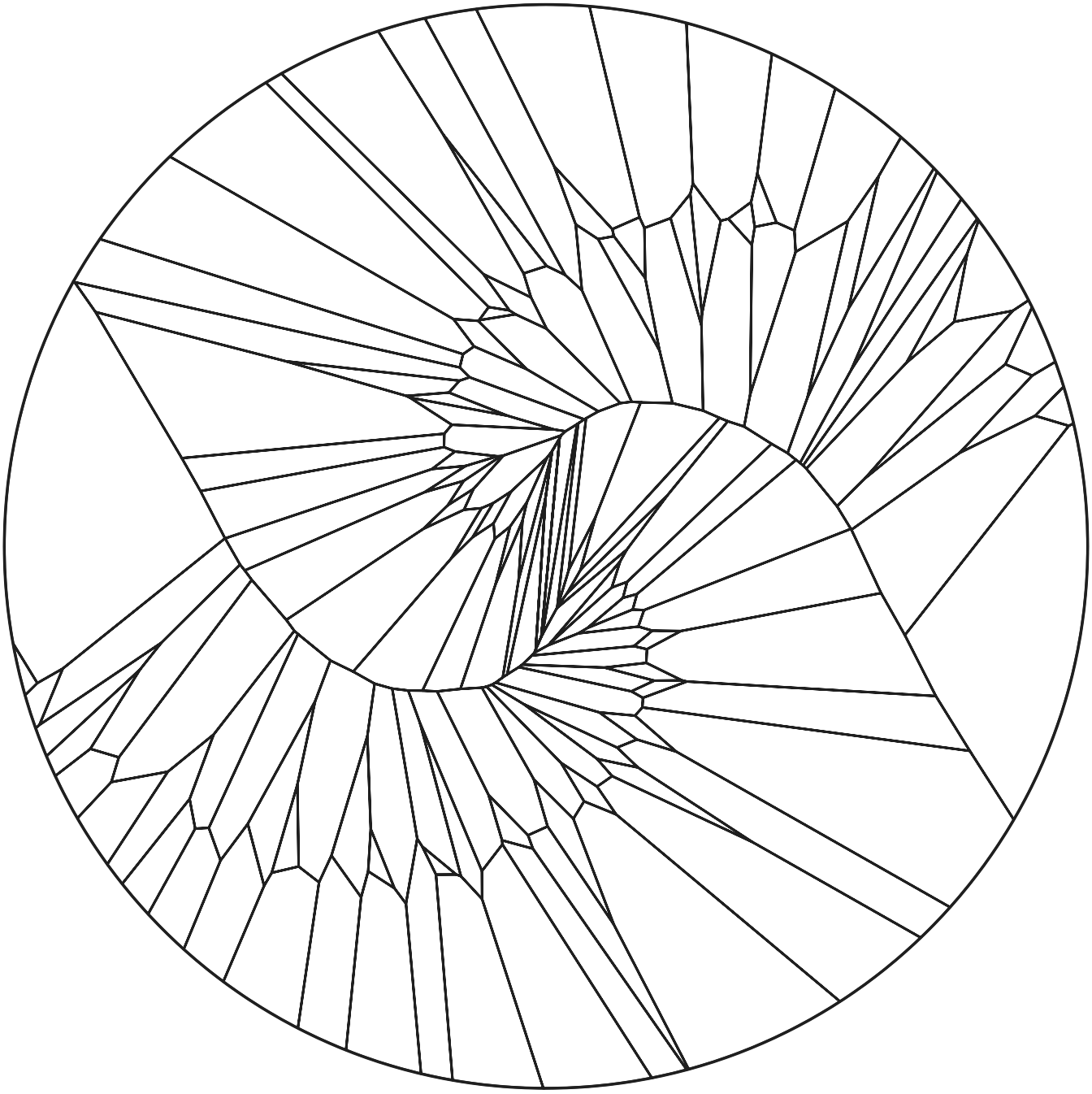
31. Chamomile Crown

Phyllotaxis Voronoi · golden angle 137.5°



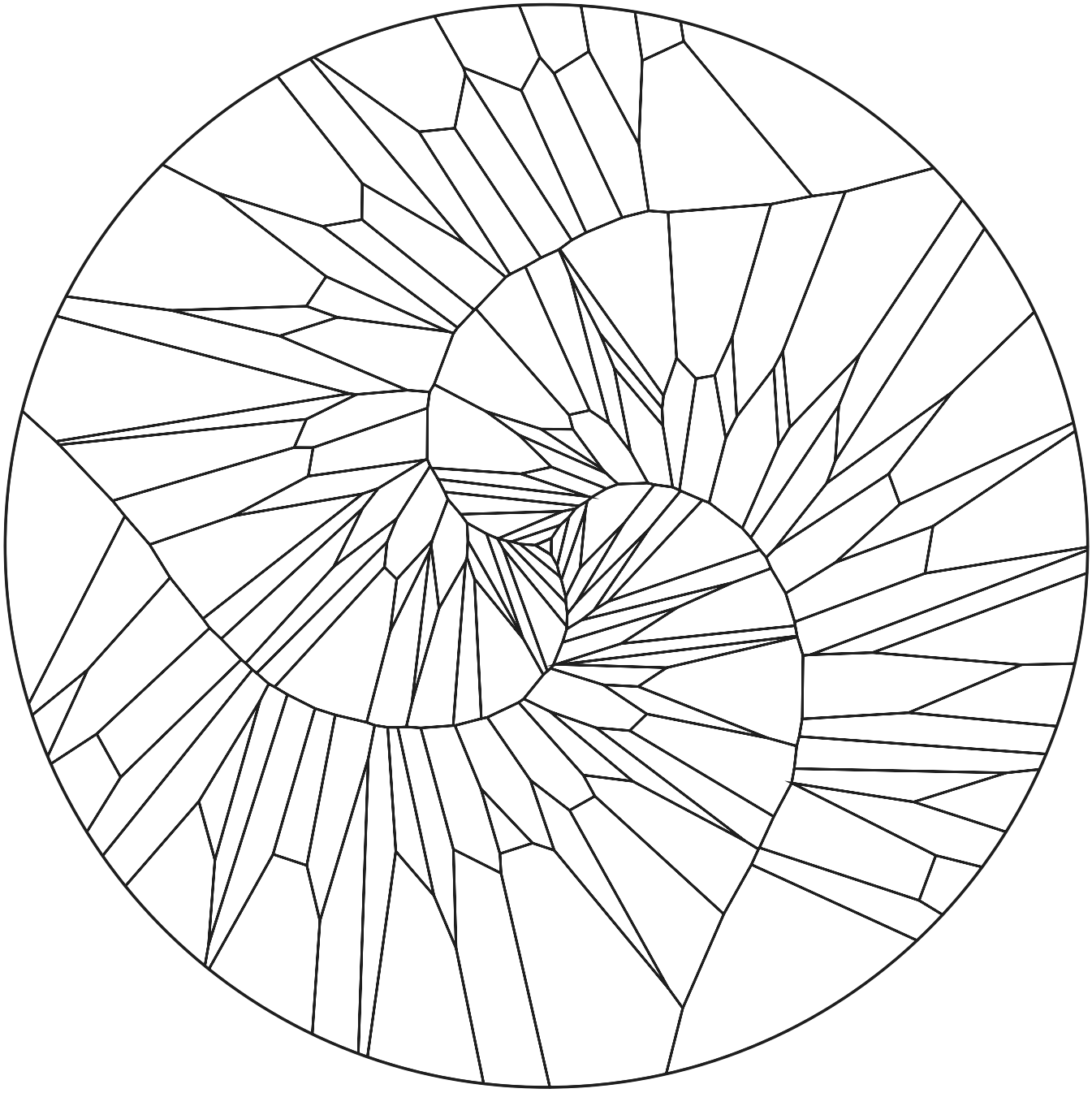
32. Helianthus

Phyllotaxis Voronoi · golden angle 137.5°



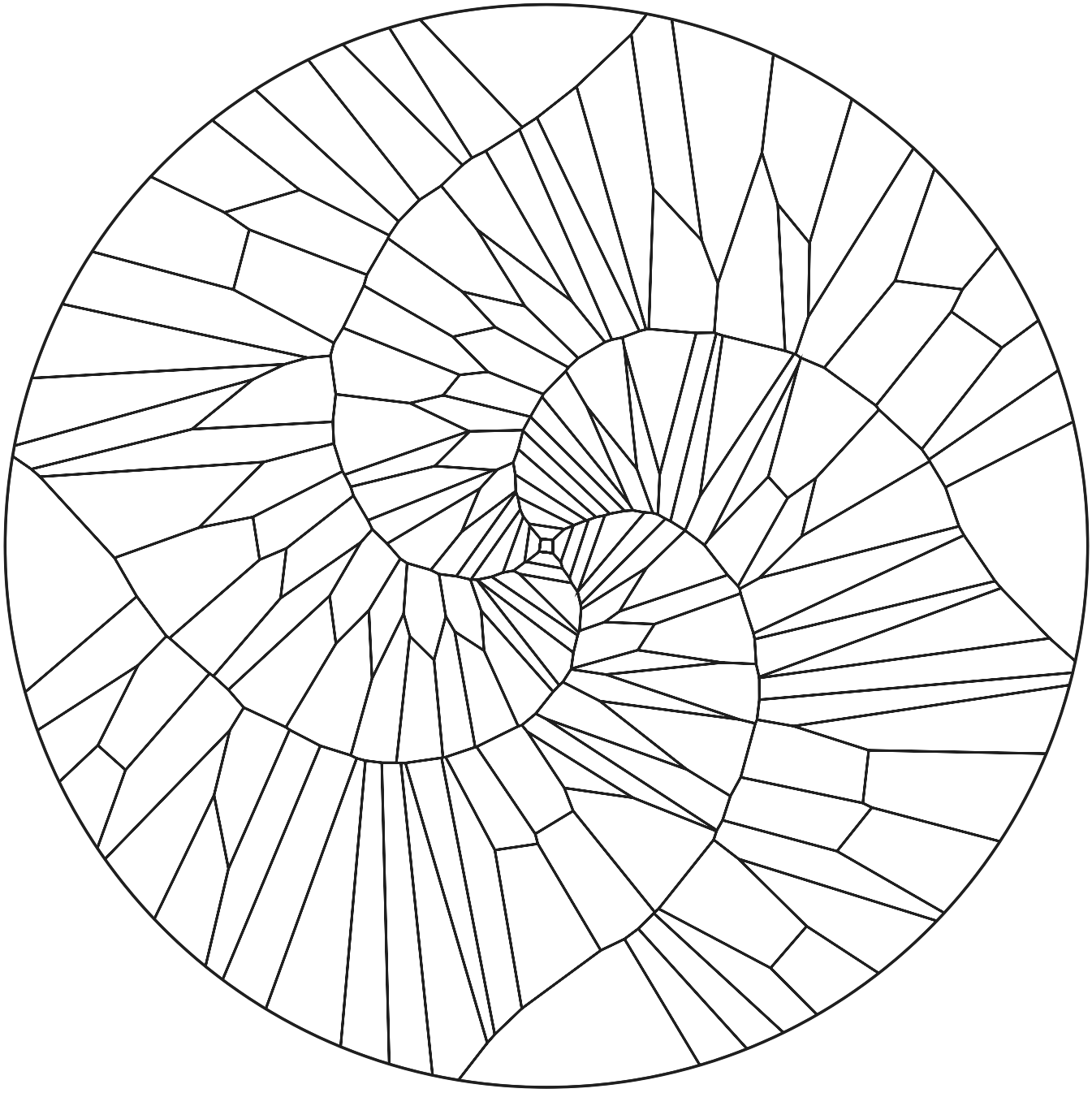
33. Spiral Galaxy

Logarithmic-spiral Voronoi · 2 arms



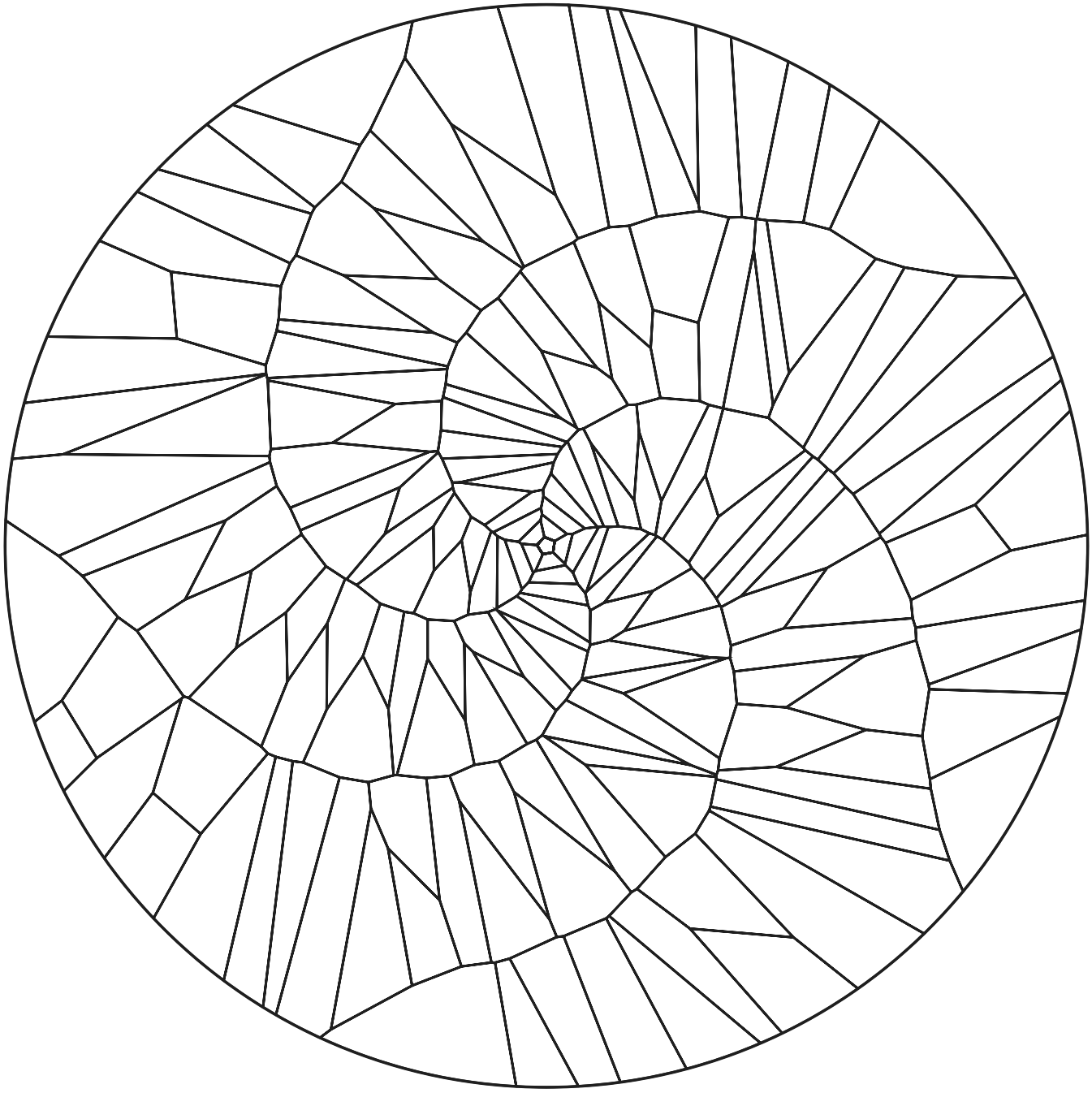
34. Triple Vortex

Logarithmic-spiral Voronoi · 3 arms



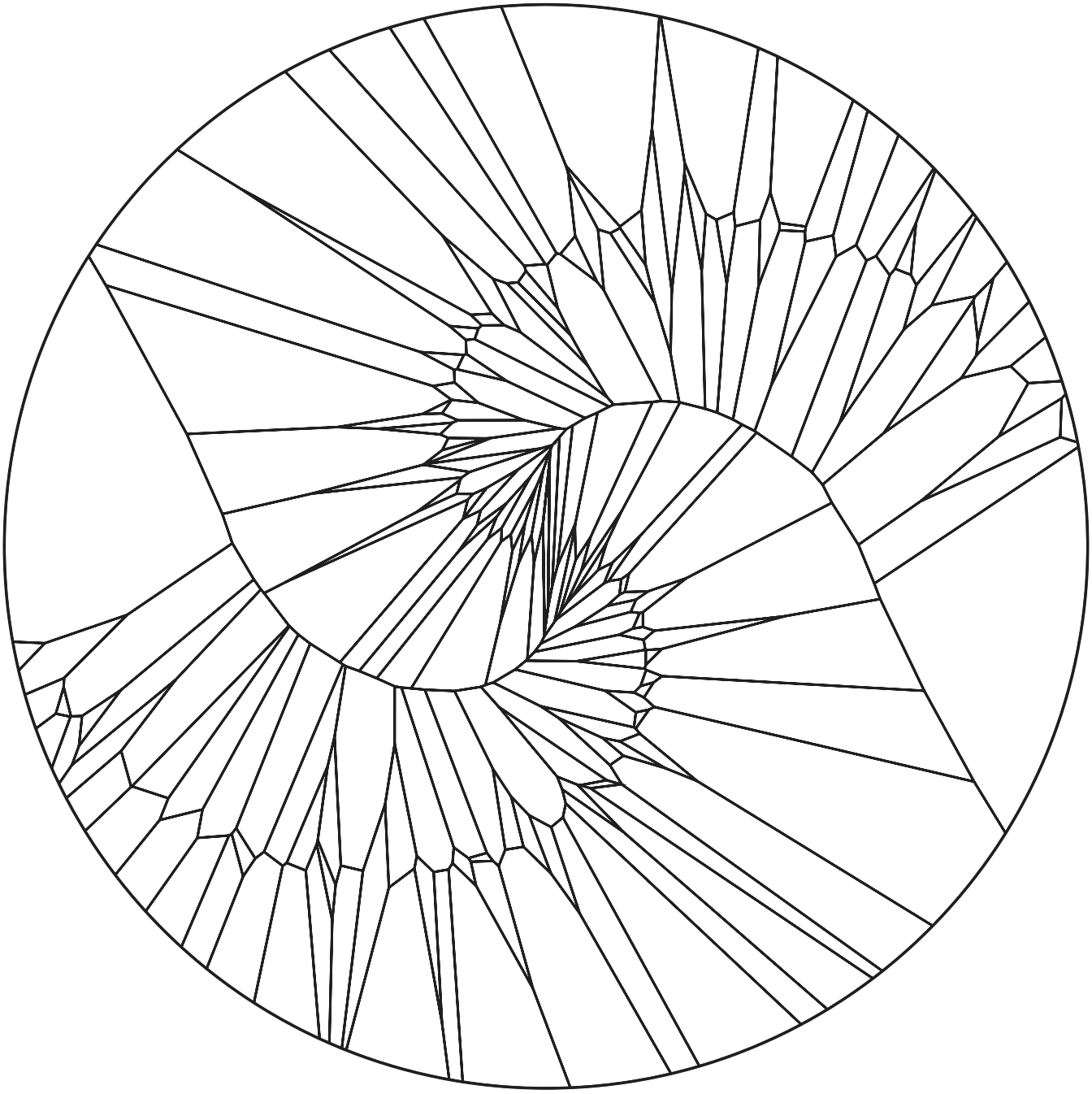
35. Pinwheel

Logarithmic-spiral Voronoi · 4 arms



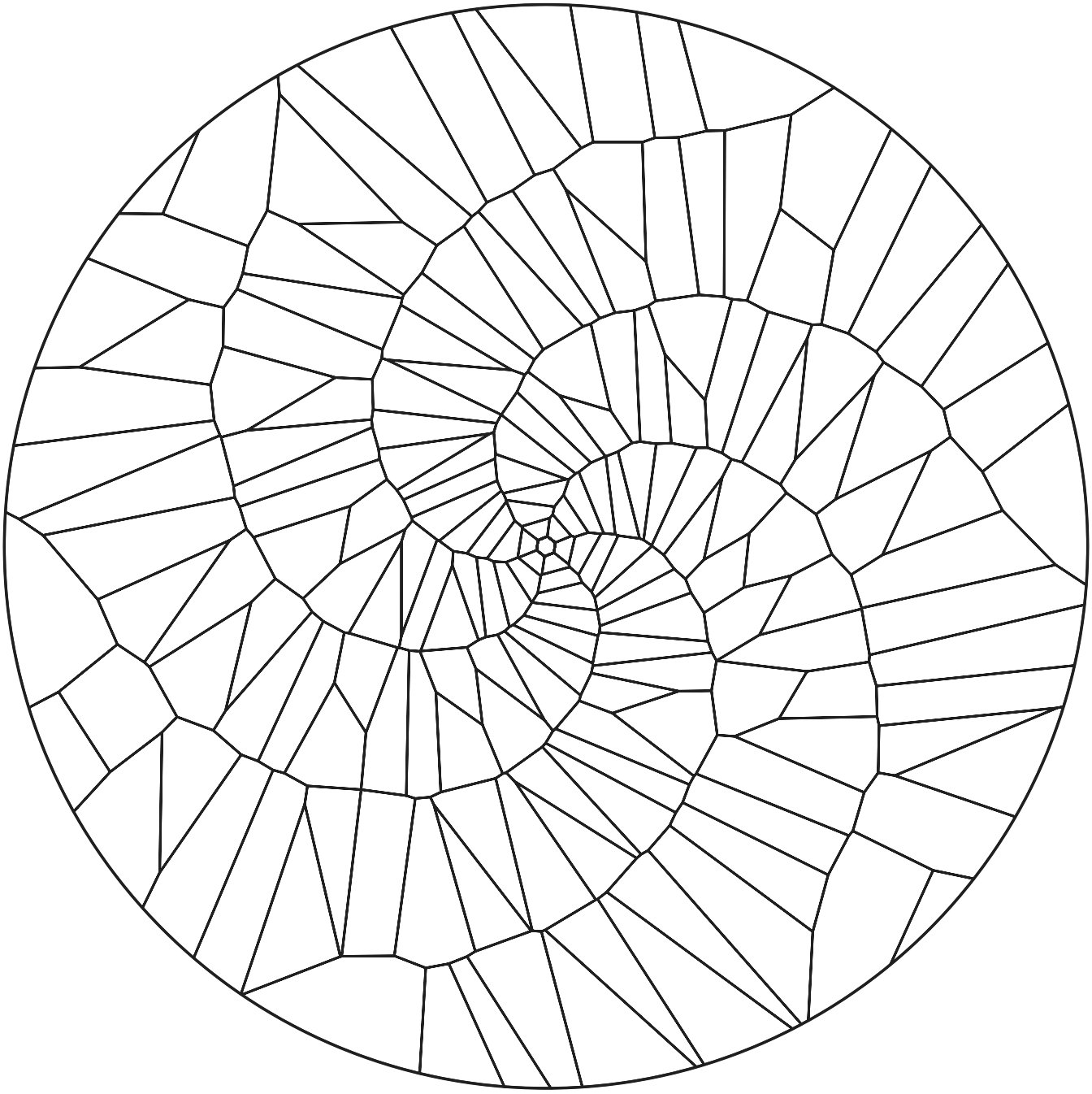
36. Catherine Wheel

Logarithmic-spiral Voronoi · 5 arms



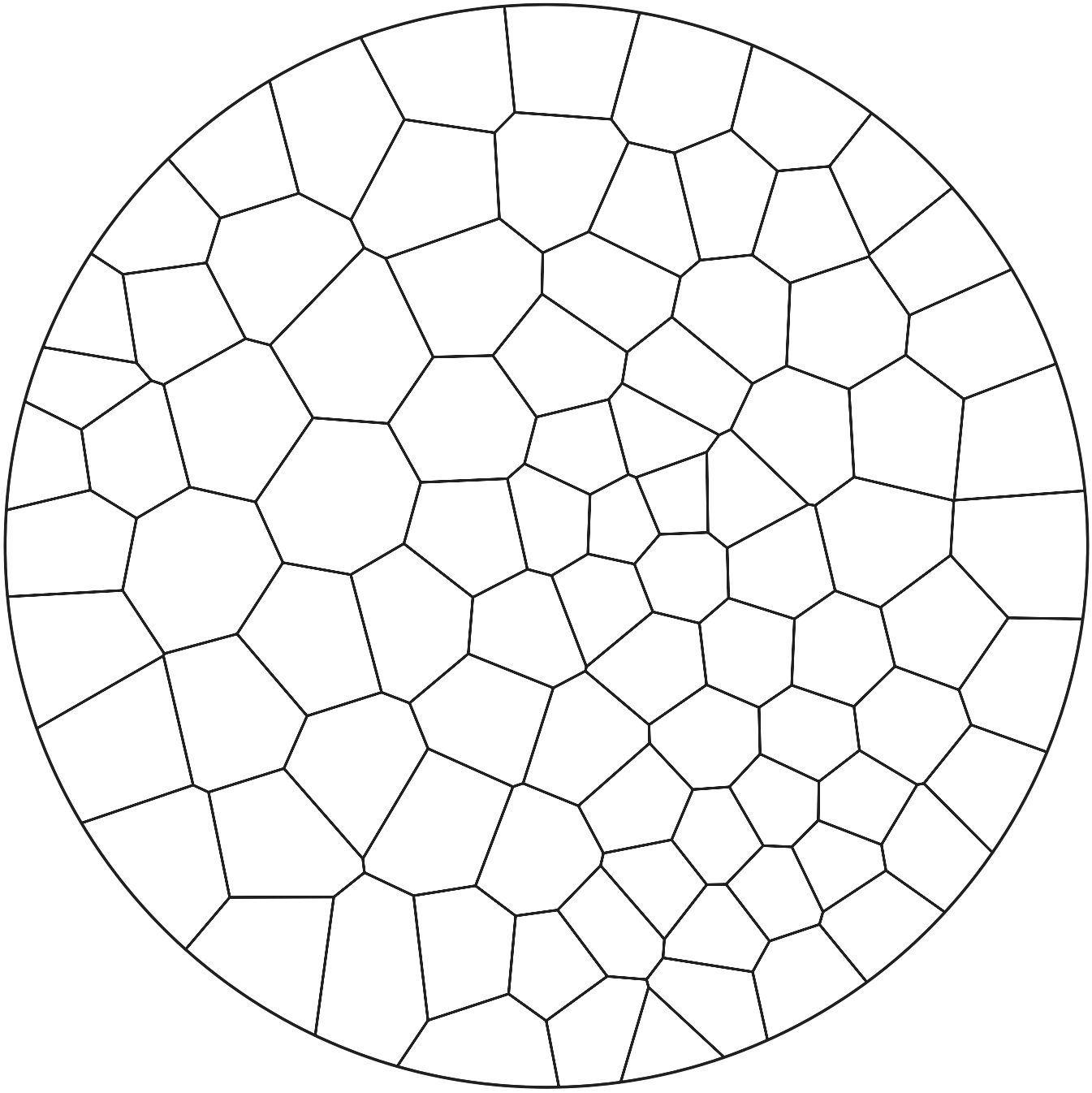
37. Whirlpool

Logarithmic-spiral Voronoi · 2 arms



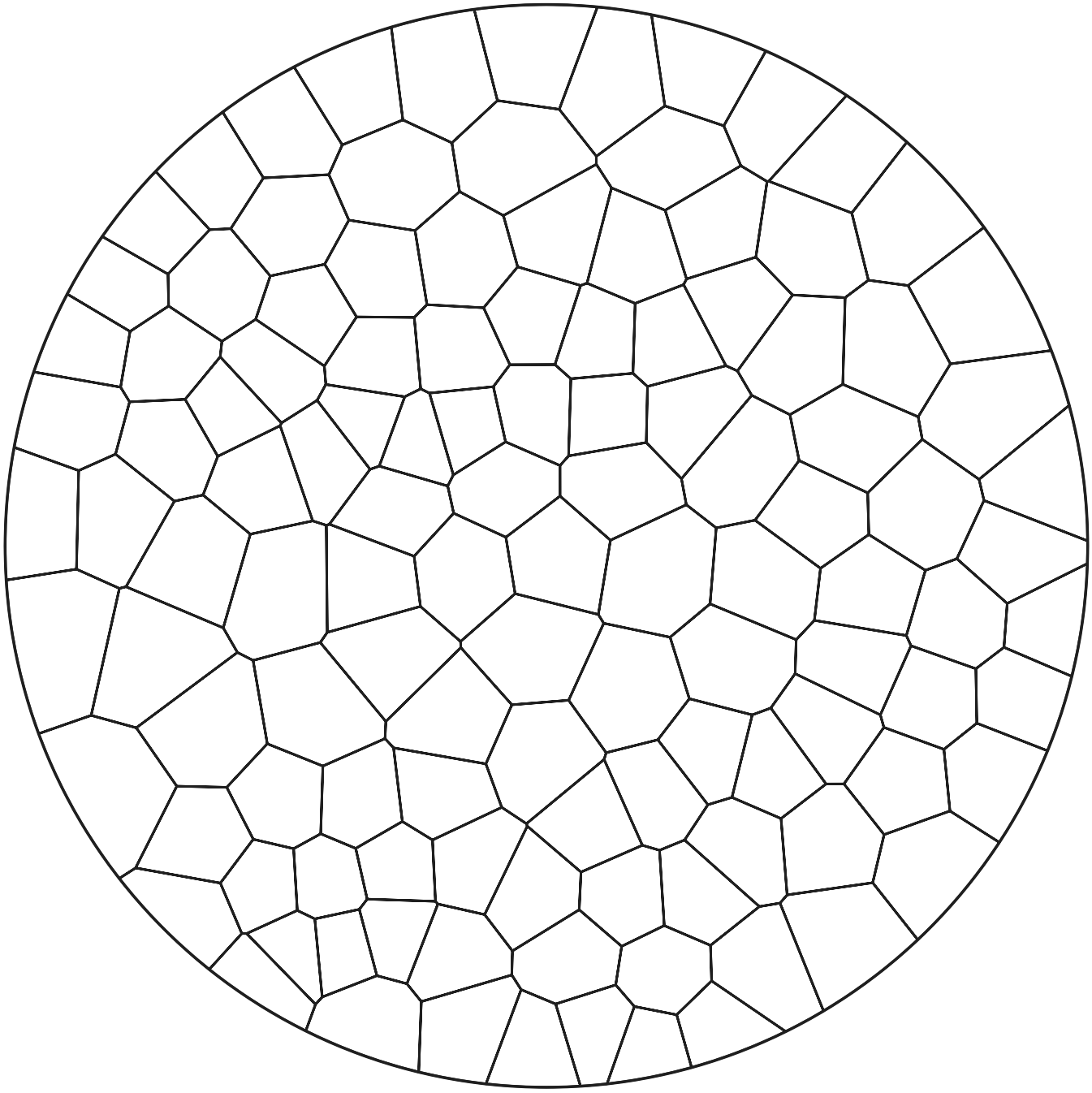
38. Nautilus Arms

Logarithmic-spiral Voronoi · 6 arms



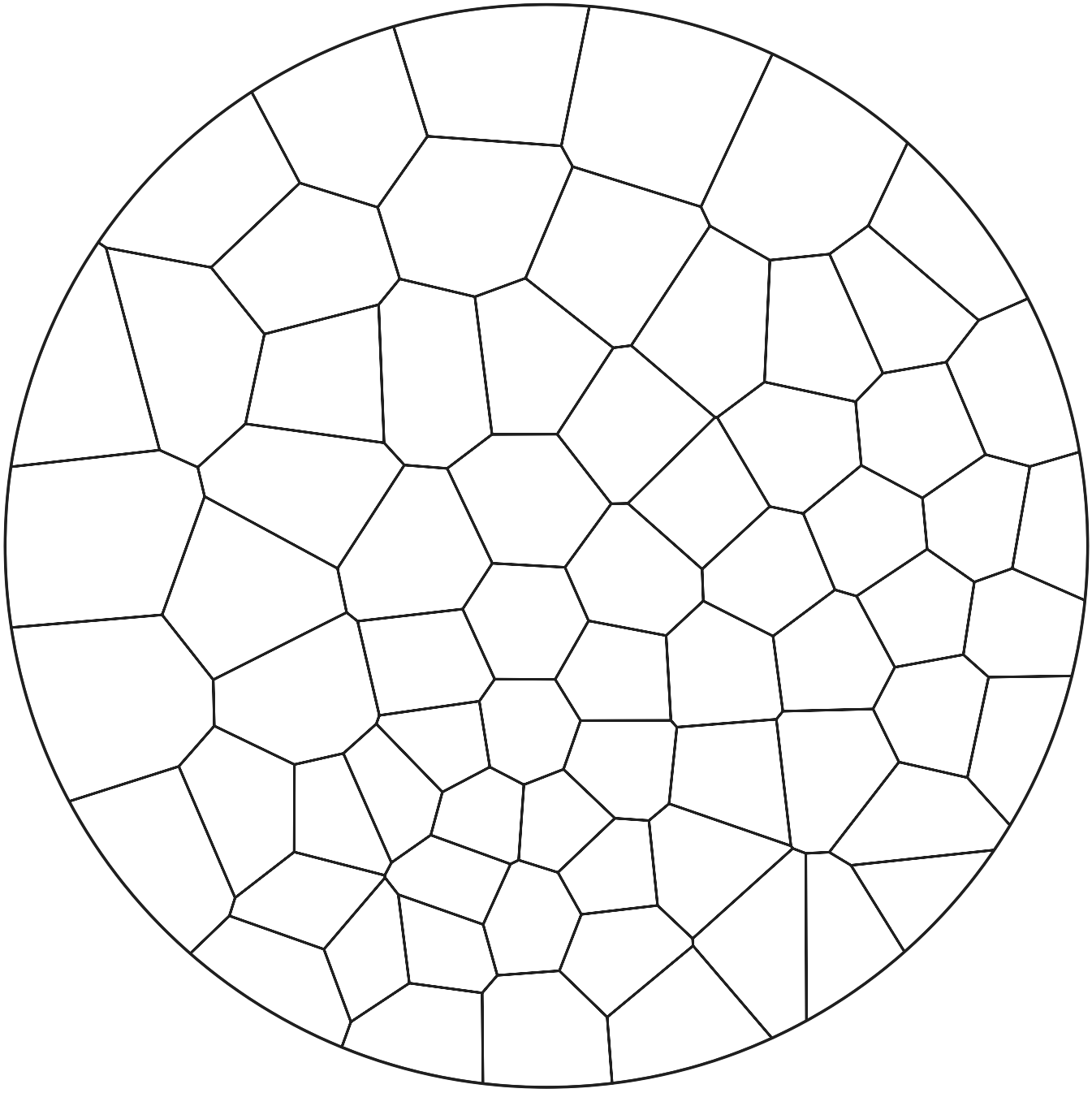
39. Cellular Garden

Lloyd-relaxed Voronoi · organic cells



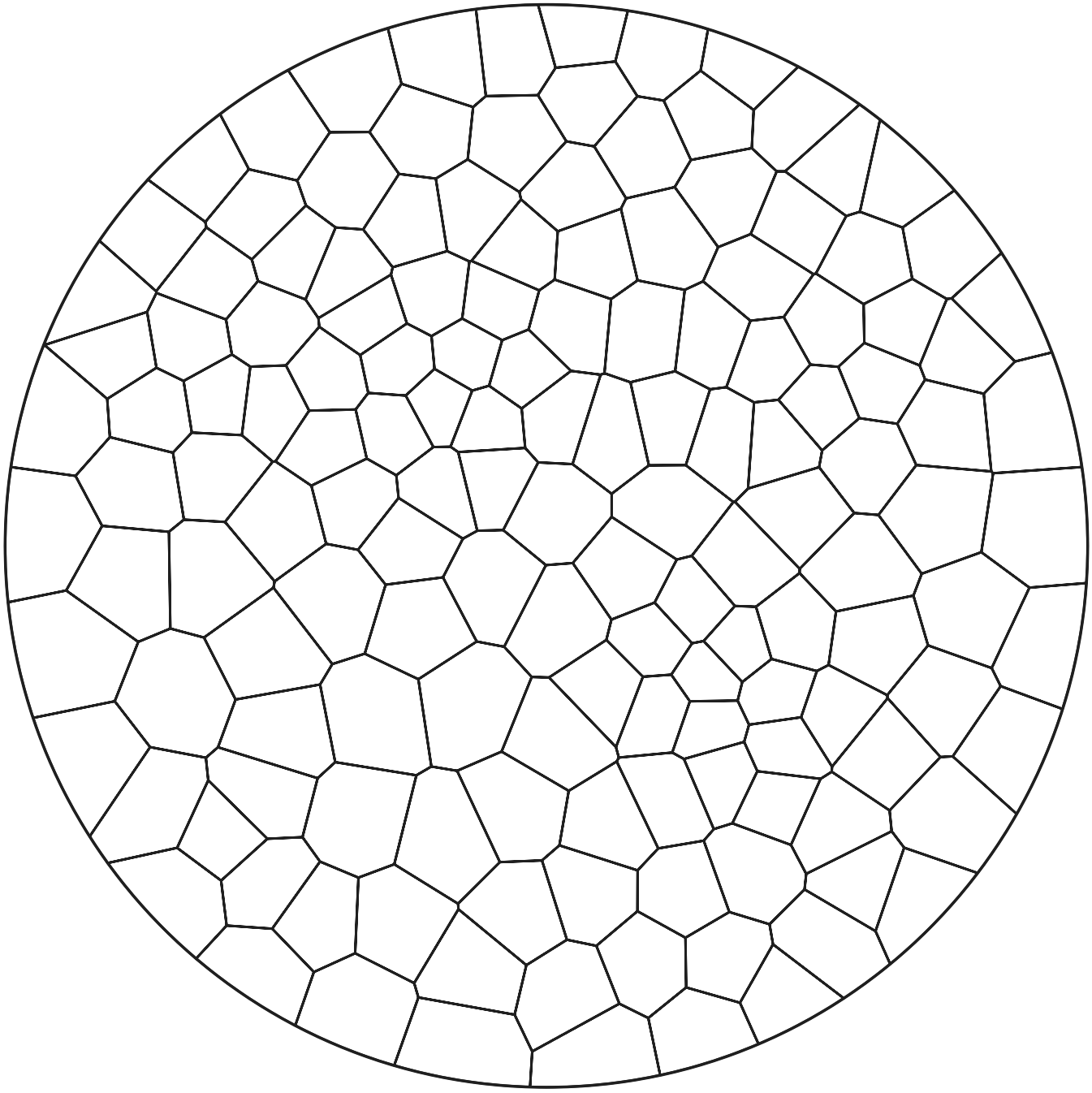
40. Honeycomb Drift

Lloyd-relaxed Voronoi · organic cells



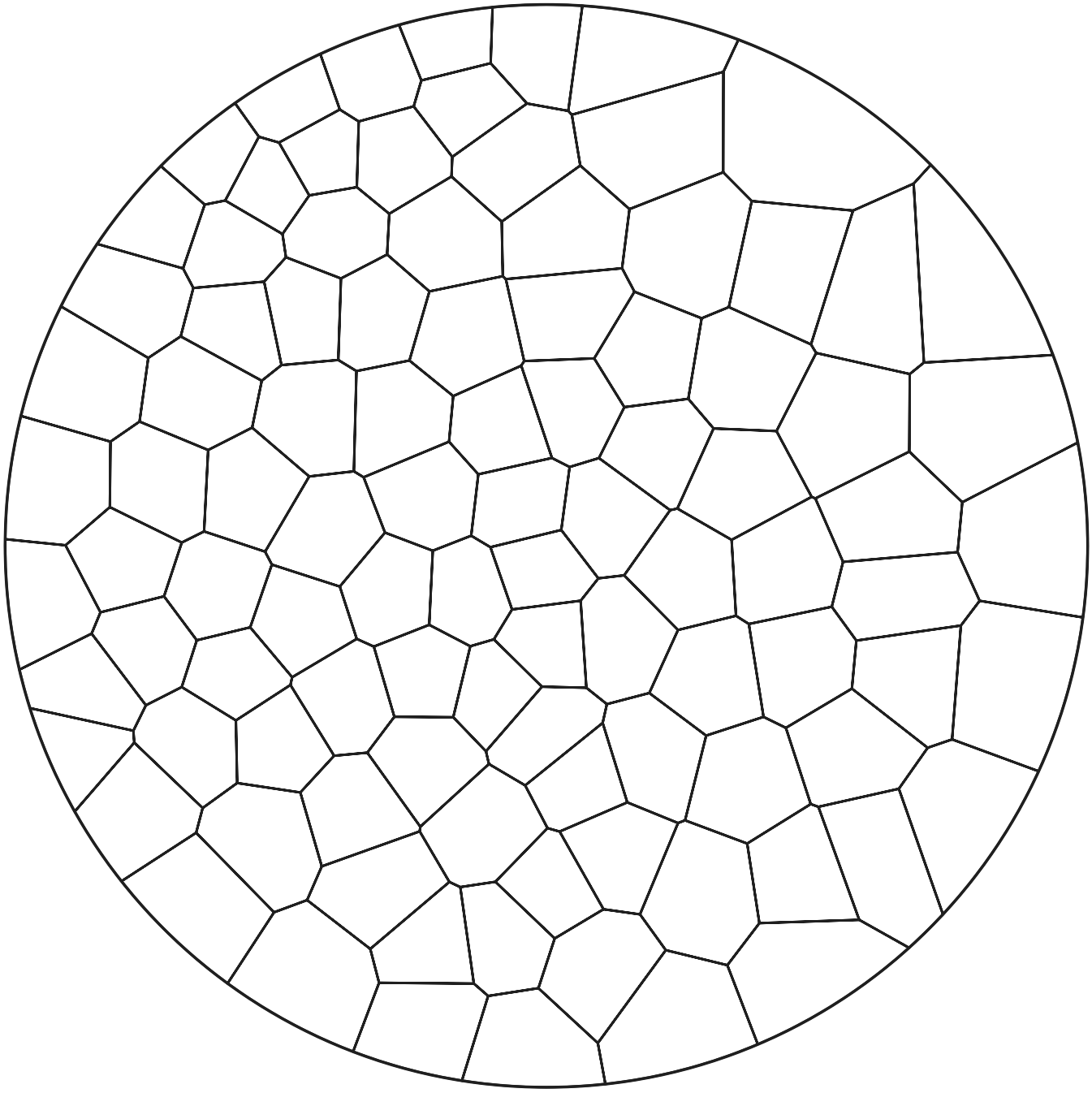
41. Soap Bubbles

Lloyd-relaxed Voronoi · organic cells



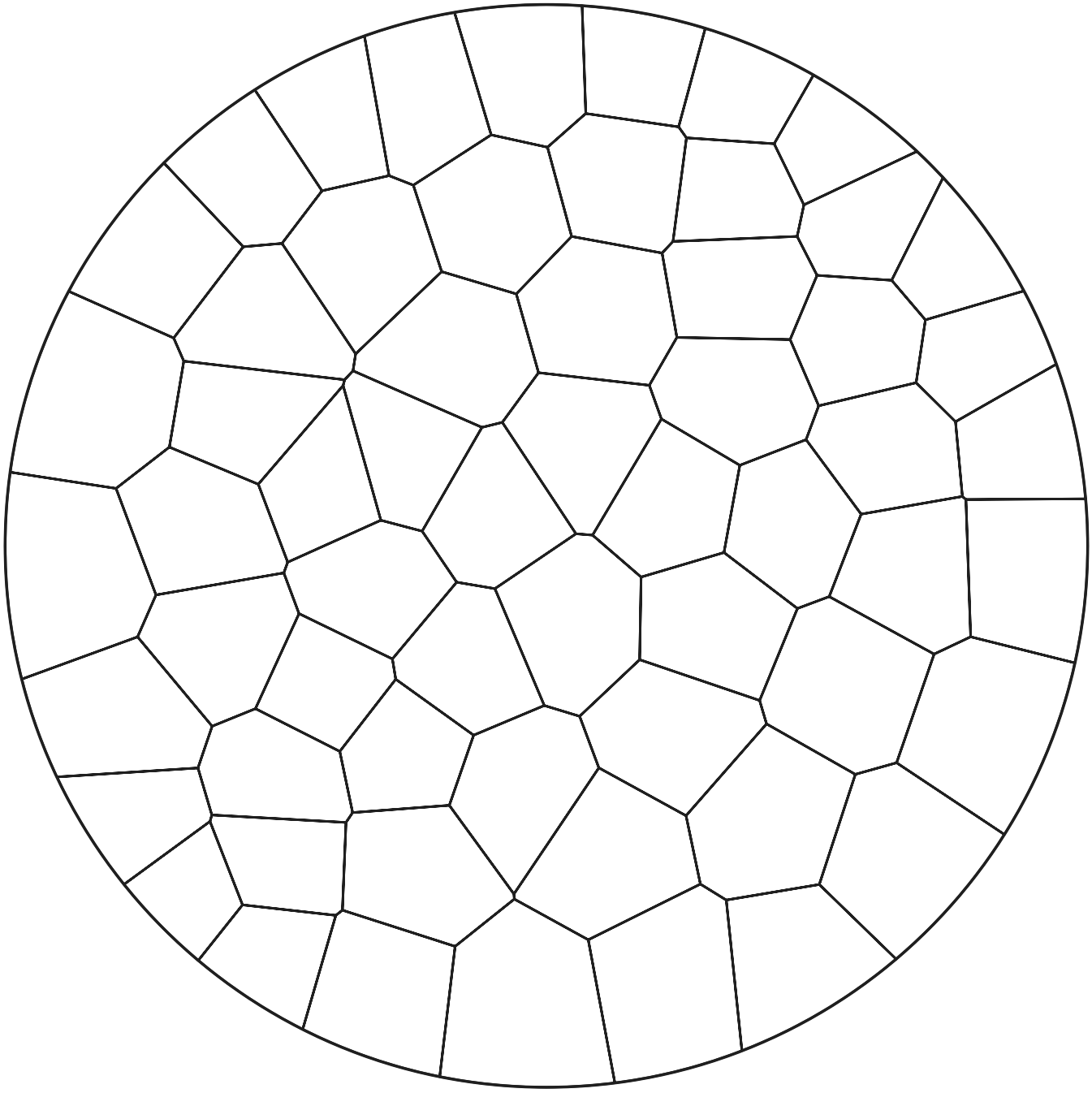
42. Leaf Venation

Lloyd-relaxed Voronoi · organic cells



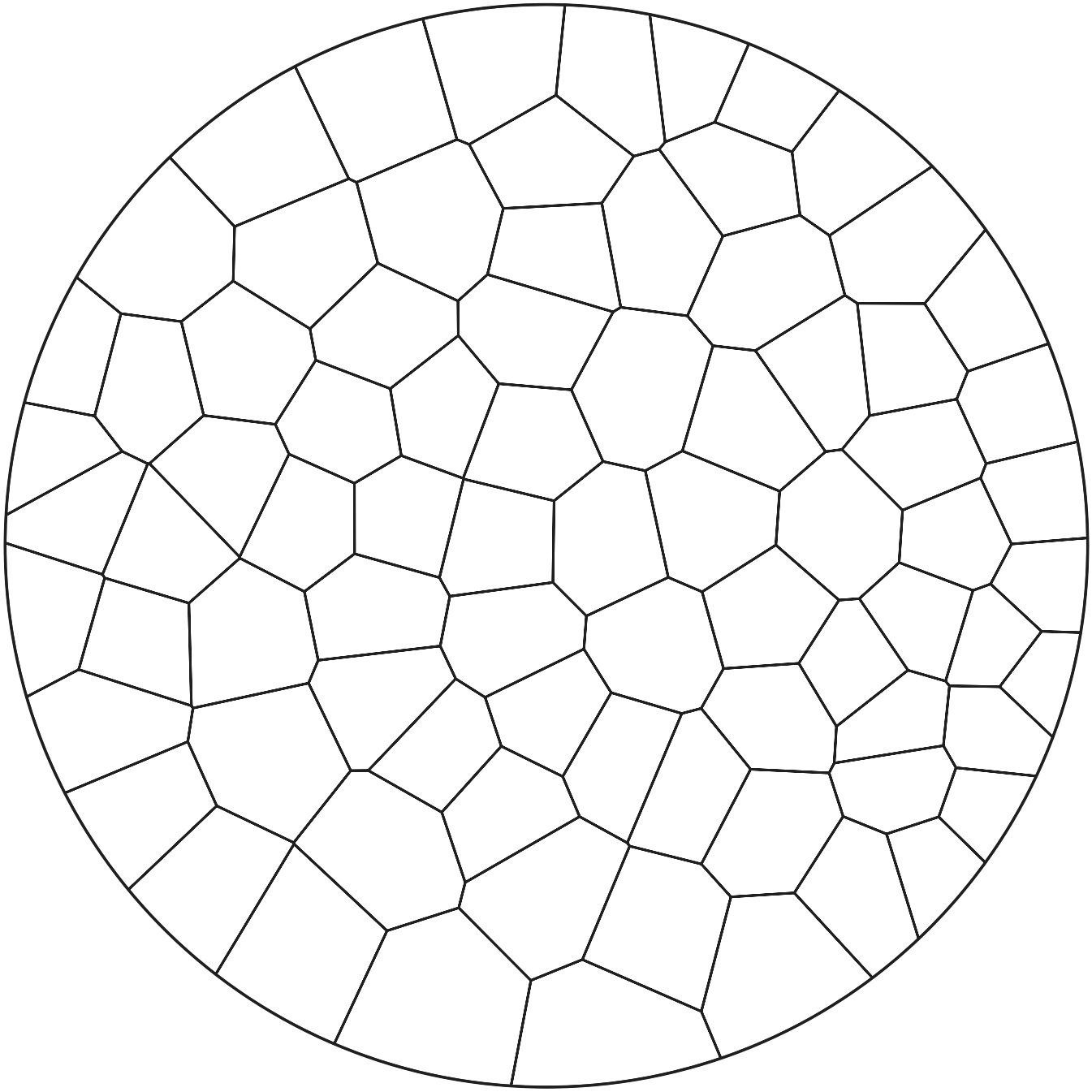
43. Dragonfly Wing

Lloyd-relaxed Voronoi · organic cells



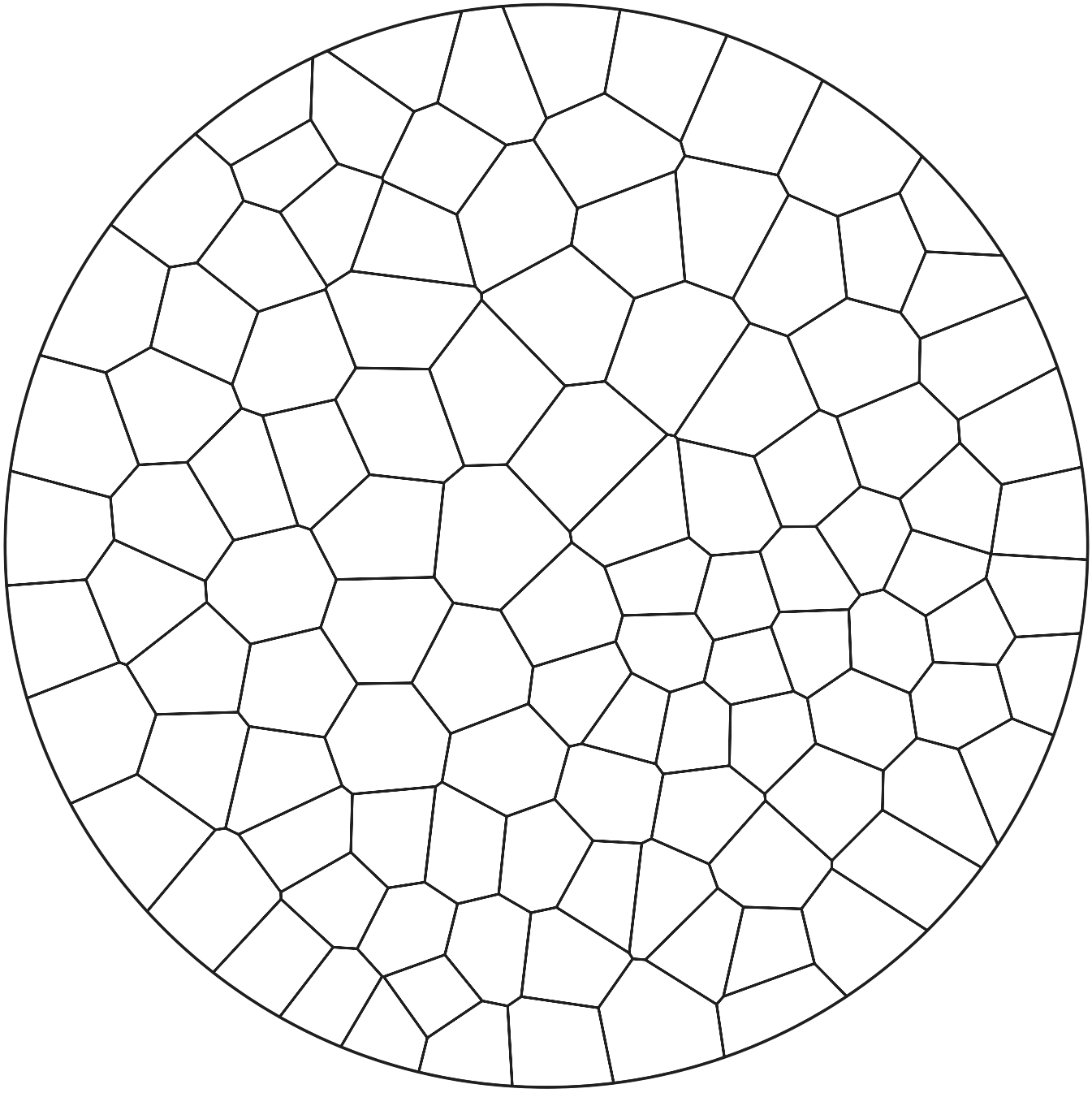
44. Giraffe Coat

Lloyd-relaxed Voronoi · organic cells



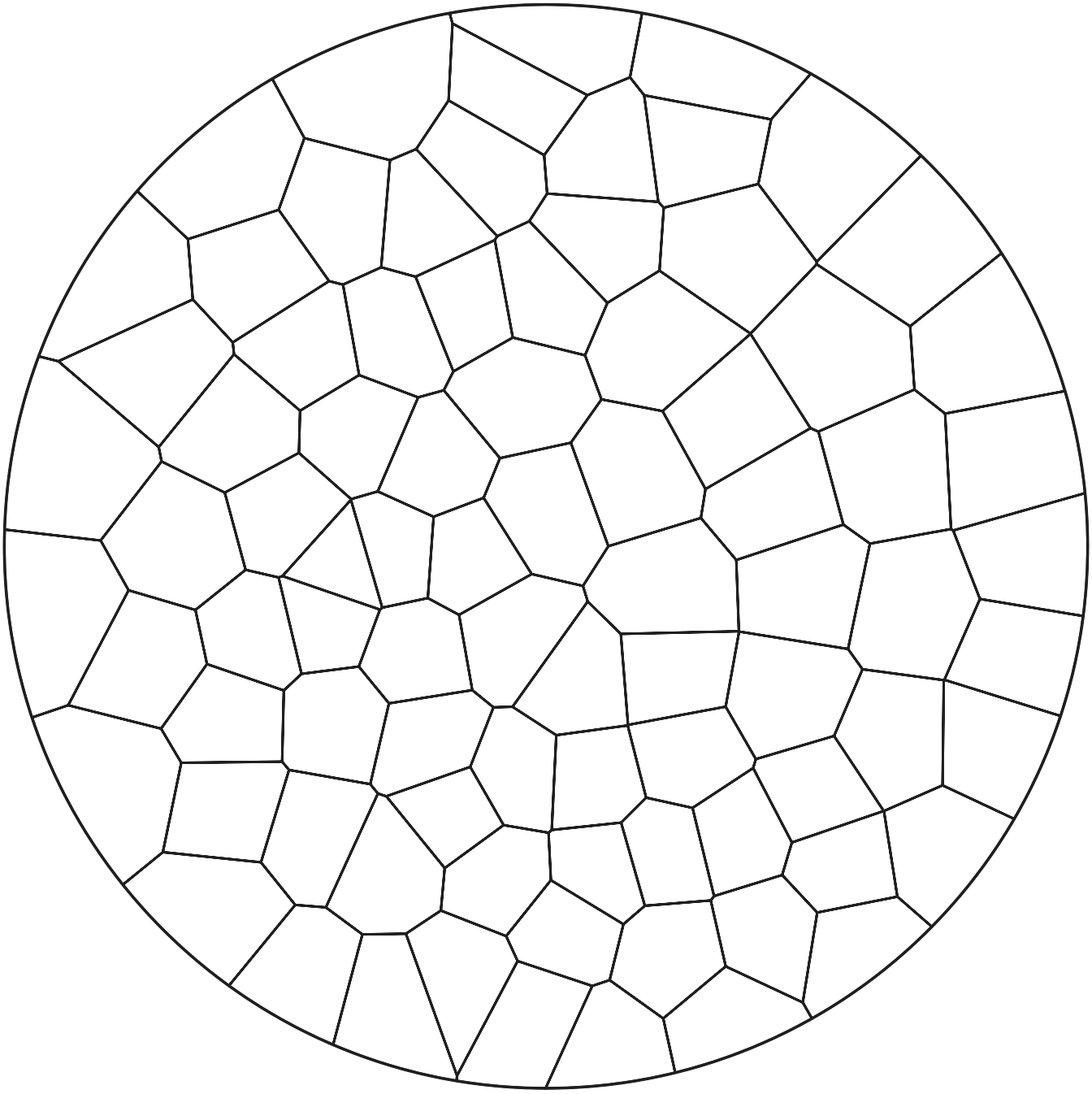
45. Cracked Earth

Lloyd-relaxed Voronoi · organic cells



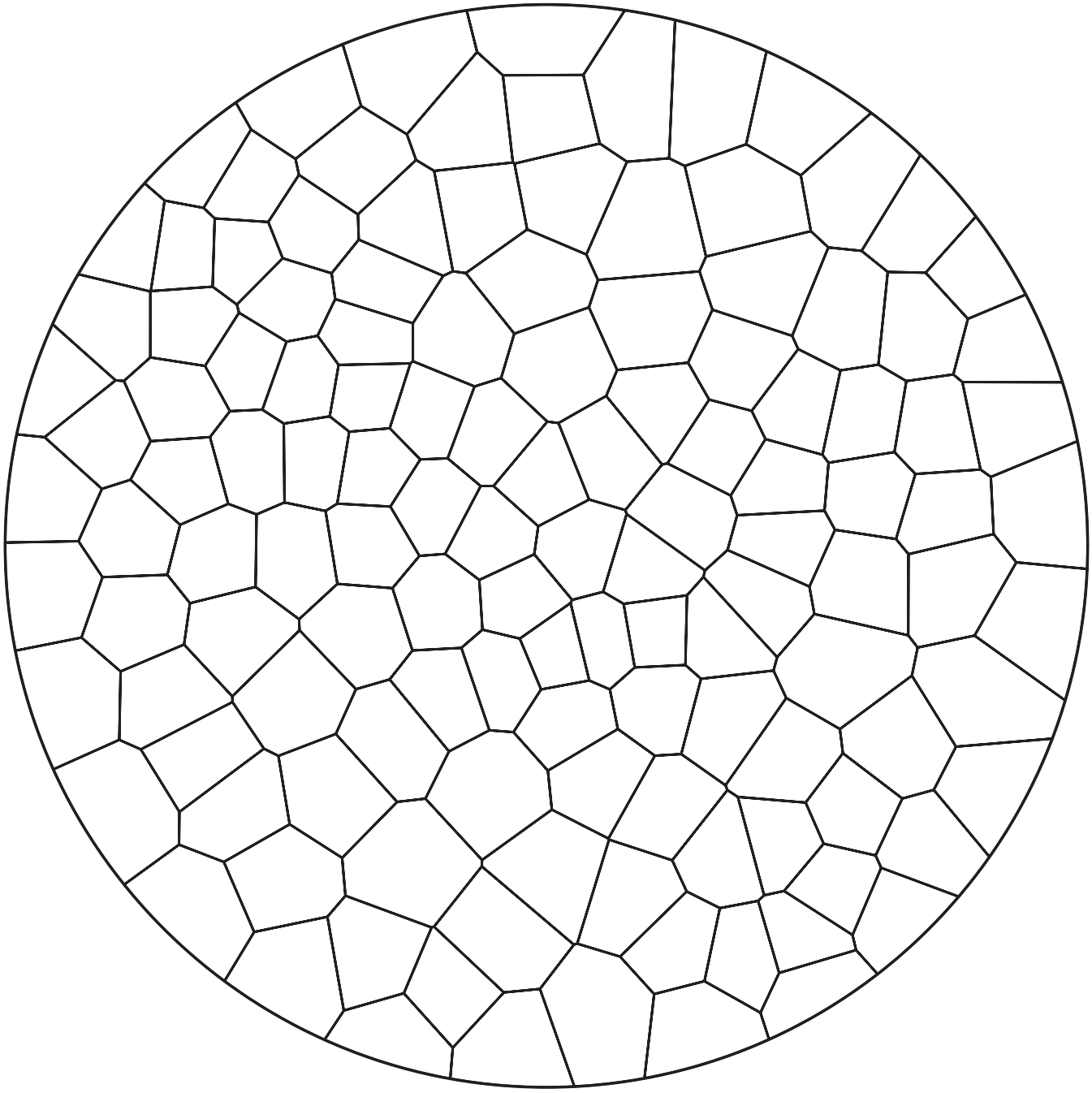
46. Stone Mosaic

Lloyd-relaxed Voronoi · organic cells



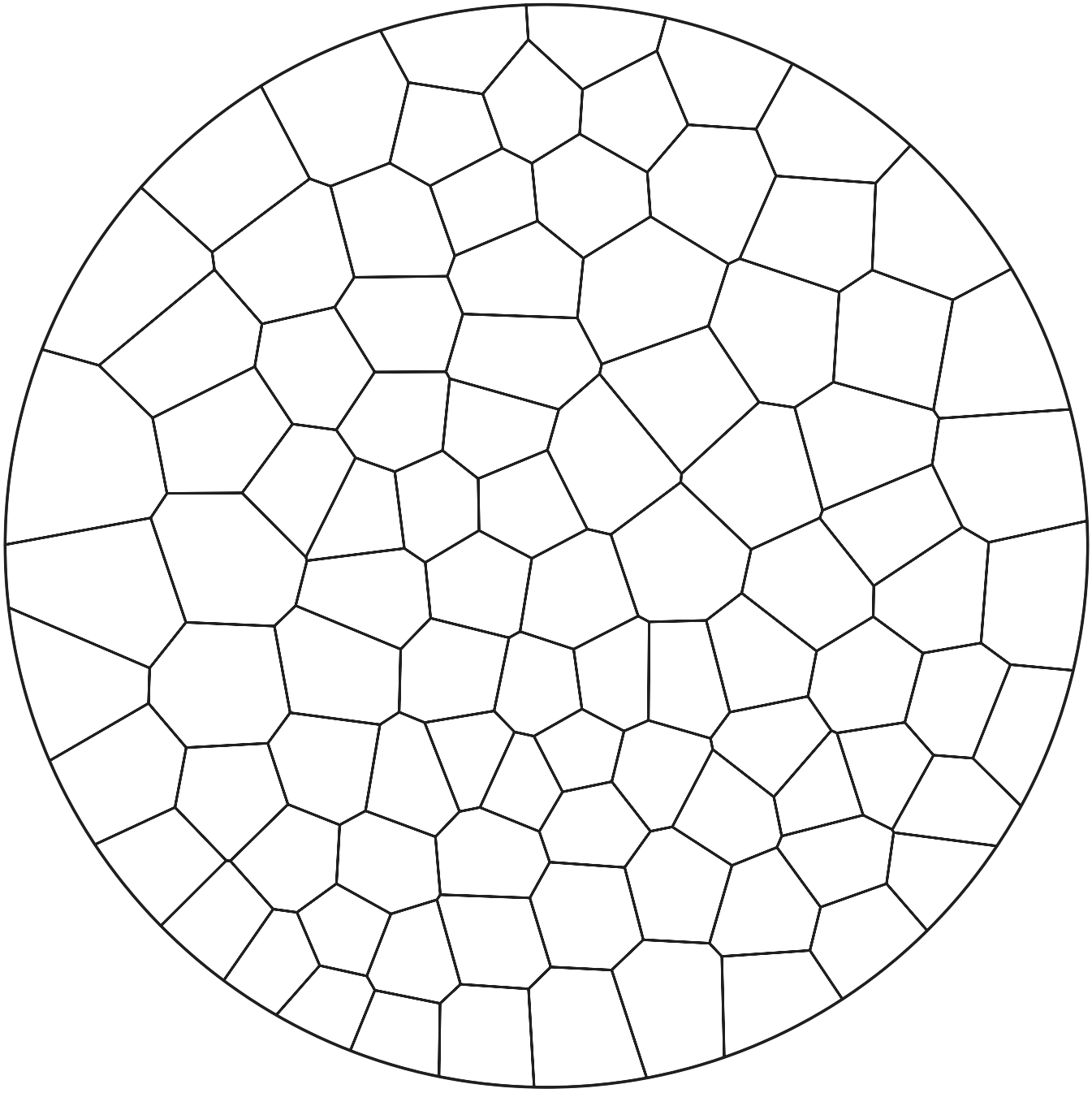
47. Tortoise Shell

Lloyd-relaxed Voronoi · organic cells



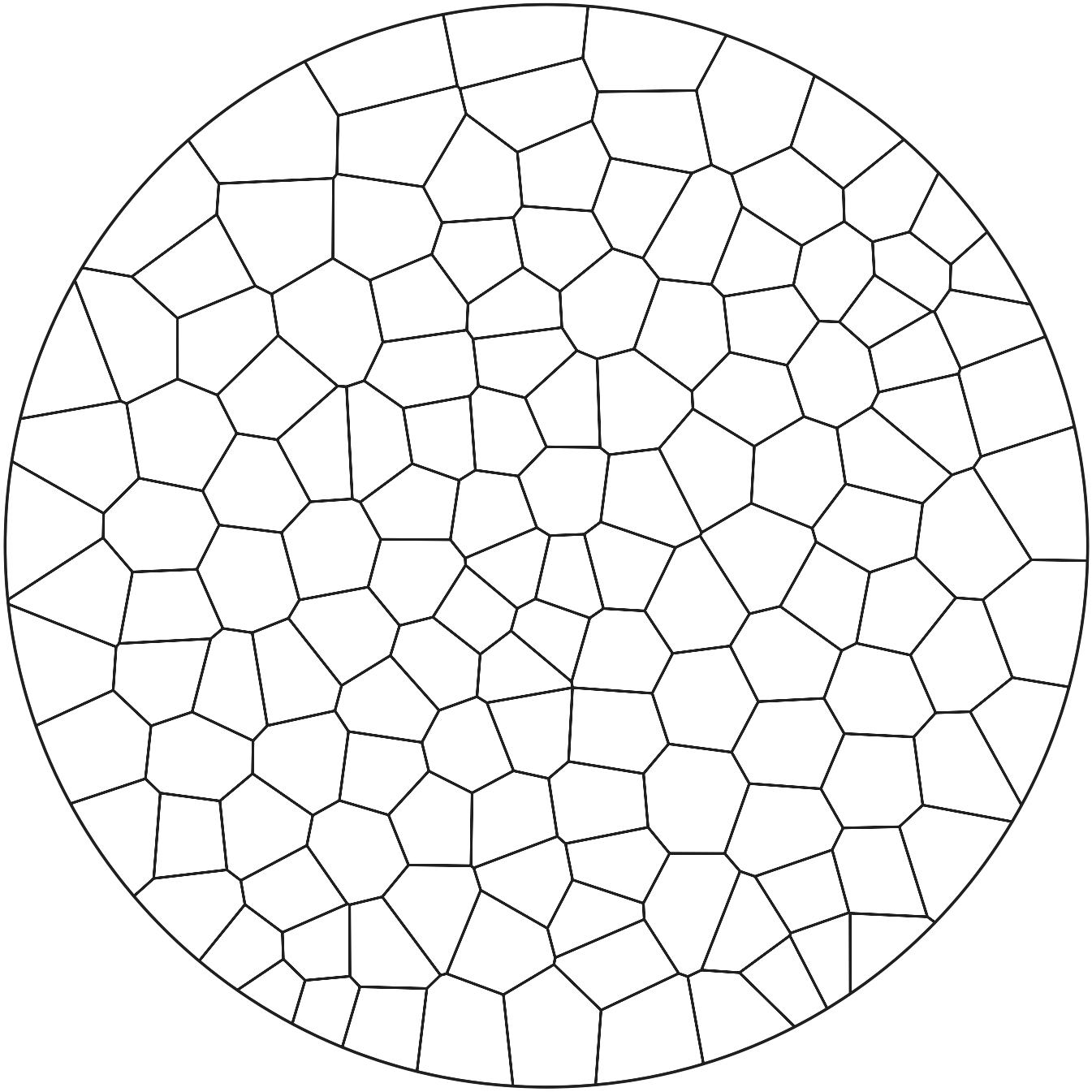
48. Coral Polyps

Lloyd-relaxed Voronoi · organic cells



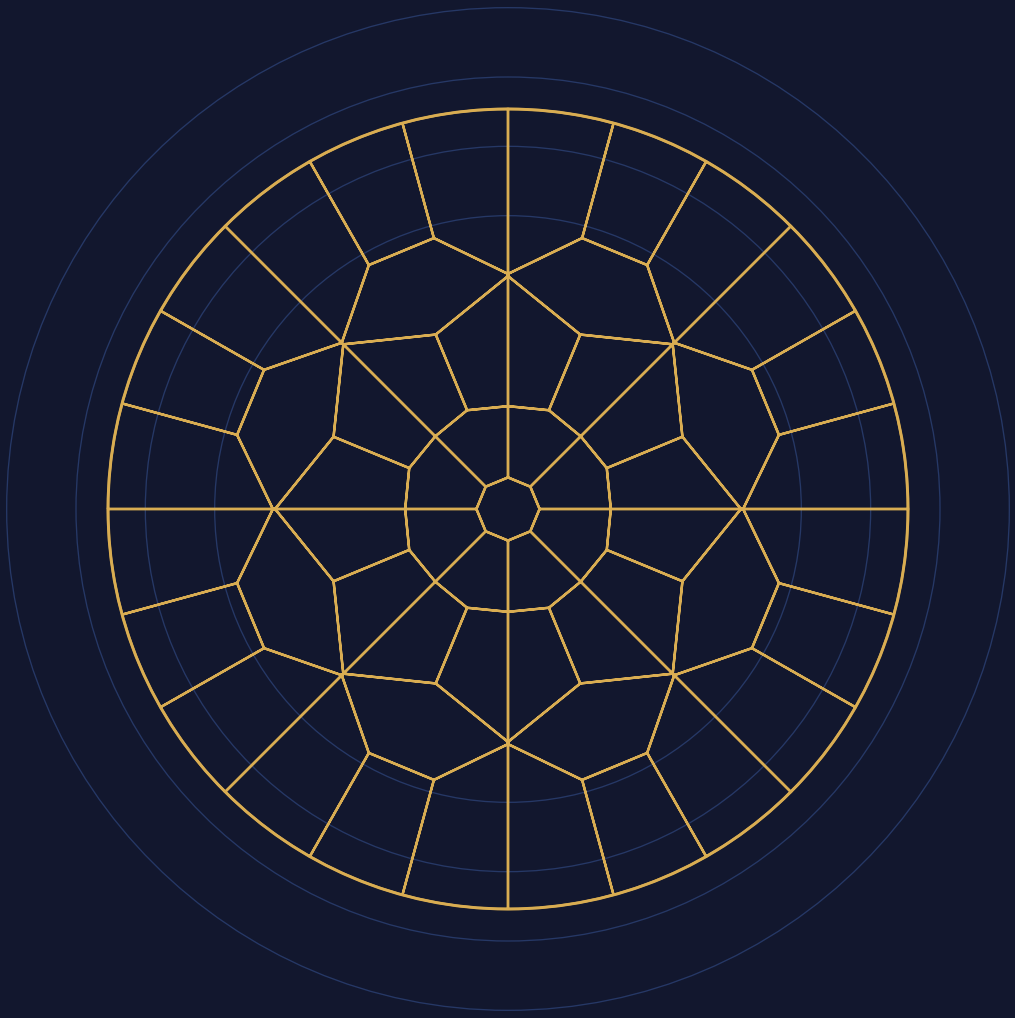
49. River Delta

Lloyd-relaxed Voronoi · organic cells



50. Frost Pane

Lloyd-relaxed Voronoi · organic cells



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Thank you for coloring!

We hope you enjoyed every cell and seed.

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